

College of Computing Sciences & IT

Study & Evaluation Scheme

Of

Bachelor of Computer Applications (B.C.A.)

[Applicable w.e.f. Academic Session 2024-25]

[As per the guidelines of UGC and National Education Policy-2020]



Accredited with NAAC **A** Grade

12-B Status from UGC

Teerthanker Mahaveer University

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TEERTHANKER MAHAVEER UNIVERSITY

Study & Evaluation Scheme				
SUMMARY				
Programme	Certificate in Computer Applications			
	Diploma in Computer Applications			
	Bachelor of Computer Applications (BCA)			
	Bachelor of Computer Applications with Research (BCA –H)			
Duration	Certificate in Computer Applications – After completion of 1 st Year.			
	Diploma in Computer Applications - After Completion of 2 nd Year.			
	Bachelor of Computer Applications (BCA) - After Completion of 3 rd Year.			
	Bachelor of Computer Applications (Honors) (BCA – H) - After Completion of 4 th Year.			
Medium	English			
Minimum Required Attendance	75%			
Credits				
Minimum Credits Required 1 Year Certificate in Bachelor of Computer Applications(BCA)		44		
Minimum Credits Required 2 Years Diploma in Bachelor of Computer Applications(BCA)		92		
Minimum Credits Required 3 Years Bachelor in Business Administration (BCOM)		144		
Minimum Credits Required 4 Years Bachelor in Bachelor of Computer Applications(BCA) (BCA – H)		188		
Assessment:				
Evaluation		Internal	External	Total
Theory		40	60	100
Practical		50	50	100
ClassTest-1 (20 Marks)	ClassTest-2 (20 Marks)	Assignment(s)	Attendance &Participation	Total
Better out of Two CTs				
20		10	10	40
DurationofExamination		External	Internal	
		3 Hours	1Hour	
To qualify the course a student is required to secure a minimum of 45% marks in aggregate including the semester end examination and teachers continuous evaluation.(i.e. both internal and external).A candidate who secures less than 45% of marks in a course shall be deemed to have failed in that course. The student should have at least 45% marks in aggregate to qualify the program				
Question Paper Structure				
1	The question paper shall have two section – Section A and Section B . The examiner shall set questions specific to respective sections. Section wise details are as under mentioned:			
2	Section A: It shall consist of short answer type questions (approx. 50 words). This section will essentially assess CO’s related to lower order thinking skills (Remembering & Understanding) . It will contain five questions with at least one question from each unit with internal choice having “or” option with optional question from the same unit. Each question shall have equal weightage of two marks and total weightage of this section shall be ten marks .			
3	Section B shall comprise of Long answer type questions (approx. 350 – 400 words). This section shall specify the higher order thinking as well as lower order thinking skills (Analyzing, Applying, Evaluating & Creating or Remembering & Understanding) to be assessed and mapped with the course outcomes stated. It shall contain five questions with at least one question from each unit with an internal choice having “or” option with optional question from the same unit. Each question shall have equal weightage of ten marks and total weightage of ten marks and total weightage of this section shall be fifty marks .			

4	Note: In case where the course content does not have the scope of assessing higher order thinking skills, questions may be framed to assess the lower order thinking skills as per the course outcomes stated.
IMPORTANT NOTES:	
1	The purpose of examination will be to assess the Course Outcomes (CO) that will ultimately lead to assessment of attainment of Program Specific Outcomes (PSO). A question paper must assess the following aspects of learning: Remembering, Understanding, Applying, Analyzing, and Evaluating & Creating (reference to Bloom's Taxonomy).
2	Case Study is essential in every question paper (wherever it is being taught as a part of pedagogy) for evaluating higher-order learning. Not all the courses might have case teaching method used as pedagogy.
3	There shall be continuous evaluation of the student and there will be a provision of fortnight progress report.
Program Structure	
A. Introduction National Education Policy-2020 aims to address many growing developmental imperatives of our country. This Policy proposes the revision and revamping of all aspects of the education structure, including its regulation and governance, to create a new system that is aligned with the aspirational goals of 21st century education, including Education for Sustainable Development Goal-SDG4, while building upon India's traditions and value systems. It lays particular emphasis on the development of the creative potential of each individual along with the development of cognitive capacities - both the 'foundational capacities' of literacy and numeracy and 'higher-order' cognitive capacities, such as critical thinking and problem solving as well as social, ethical, emotional capacities and dispositions. One of its focal points is to bring about significant changes in the Indian education system at the undergraduate level. The NEP-2020 also proposes a credit-based system for undergraduate programs that allows students to select courses from multiple disciplines and complete their degrees at their own pace. A comprehensive and integrative education is essential to lead the country into the 21st century and the fourth industrial revolution. Imaginative and flexible curricular structures will enable creative combinations of disciplines for study and would offer choice-based curriculum with multiple entry and exit points, thus, removing currently prevalent rigid boundaries and creating new possibilities for life-long learning. The Bachelor of computer Application syllabus of Teerthanker Mahaveer University is designed in compliance with the National Education Policy 2020 (NEP 2020) including a wide range of courses that aim to provide students with a well-rounded education. The program is a four-year undergraduate degree that allows students to specialize in one or more disciplines of their choice and multiple entry & exit. The curriculum is designed so as to give students an in-depth knowledge of the academic disciplines and applied functional areas necessary to meet the requirements of business enterprises and the industry. BCA is an undergraduate degree program. This programme provides in-depth knowledge about basic & higher-level knowledge in computer field or respective minor disciplines for example, Full Stack, Artificial intelligence, Data Analytics, Data Science, Cyber security, Computer Science additionally. This Honours course is an important and valuable one that provides opportunities to the students of taking some of the subjects of a Master's degree. After completing the course, they can go to many fields to obtain jobs. Foundational Courses are designed to develop critical thinking and problem-solving skills, and include subjects like mathematics, science, and languages. Core Courses cover the fundamental concepts and theories of the chosen subject(s) of study. Elective Courses (Deep Specialization) allow students to explore their interests and deepen their understanding of their chosen specialization. Students can choose courses from a range of specialization, including Full Stack, Artificial intelligence, Data Analytics, Data Science, Cyber security, Computer Science. The curriculum also has minor specialization which includes 10 groups i.e. Artificial intelligence, Data Analytics, Data Science, Cyber security. Multidisciplinary Courses integrate knowledge and methods from multiple disciplines and aim to develop a holistic understanding of complex issues. For example, a course on climate change might integrate knowledge from environmental science, corporate skill, human value, Indian knowledge system, and law. Skill-based Courses aim to develop practical skills that are relevant to the chosen field of study. Overall this syllabus is designed in compliance with the NEP-2020. The new syllabus is multidimensional, with a focus on developing skills like critical thinking, problem-solving, project based learning and communication & to prepare students for a wide range of career options and lifelong learning. This will result in significant changes; present an opportunity to create a more dynamic and relevant education system that prepares students for the challenges of the 21st century. B. Credit Framework for Undergraduate Program – The duration of BCA Four Year Degree Program having eight semesters with multiple entry and exit options.	

a. Credit Requirements for Exit Options			
Exit Options	Minimum Credits Requirements	NSQF Level	Requirement for Exit
Certificate in Commerce – After completion of 1st Year.	44+4*	5	*4 extra Credits to be earned along with 44 Credits (Minimum Credits Requirements) after completion of II semester.
Diploma in Commerce – After Completion of 2nd Year.	92+4*	6	*4 extra Credits to be earned along with 92 Credits (Minimum Credits Requirements) after completion of IV semester.
Bachelor’s Degree (BCA) – After successful completion of Third year	144	7	
Bachelor’s Degree with Honors (BCA – H) – After successful completion of fourth year	188	8	
b. National Skills Qualifications Framework (NSQF) Levels:			
Exit Option	NS QF Level	Professional Knowledge	Skill
At the end of first year	5	Knowledge of facts, principles, processes, concepts in a field of work or study	Desired Problem-Solving skills, understanding of social, political environment, and some skill in collecting and organizing information, communication
At the end of Second year	6	Factual and theoretical knowledge in the broad context with in a field of work or study	Reasonably good in arithmetic calculations, understanding social and political environment, good skill in collecting and organizing information, logical communication
At the end of Third year	7	Wide ranging factual and theoretical knowledge in the broad context with in afield of work or study	Good logical and mathematical skill, understanding social, political and natural environment, good in collecting and organizing information , communication and presentation skills
At the end of Fourth year	8	Comprehensive, cognitive theoretical knowledge and practical skills to develop creative solutions to abstract problem	Exercise management and supervision in the context of work /study in unpredictable changes, responsible for the development of self and others
c. Academic Bank of Credits (ABC):			
As per the National Educational Policy (NEP) 2020, the Academic Bank of Credit offers the flexibility of curriculum framework and interdisciplinary /multidisciplinary academic mobility of students across Higher Educational Institutes (HEIs) with appropriate credit transfer mechanism. In furtherance to these guidelines the Teerthanker Mahaveer Institute of Management and Technology, Teerthanker Mahaveer University, Moradabad has designed a four years undergraduate program. As a pre-requisite a student’s /learners should register themselves in the website of Academic Bank of Credit. The credits earned by the student /learner will be stored in it. A Student/Learner would be required to complete the course as per the ABC (Academic Bank Credit) policy of UGC. The validity of the credits earned for a course is seven years only. Credit earned cannot be used for multiple times / purpose. The credit once use cannot be utilize again. .			
C. Curriculum Framework:			
Curriculum lay emphasis on the implementation of New Education Policy 2020 (NEP-2020). The courses are broadly classified as follows: Discipline Core Course (CC), Discipline Major Course, Discipline Minor Course, Multidisciplinary Course, Skill Enhancement Courses (SEC), Ability Enhancement Courses (AEC) and Value Added Courses (VAC). The curriculum also gives emphasis on Summer Internship Report, Major Project Report, Industry Project Report, Research Project Report.			

I. Discipline Core Courses:

Major Core Course is the discipline or subject of main focus and the degree will be awarded in that discipline. Students should secure the prescribed number of credits (about 50% of total credits) through core courses in the major discipline and Laboratory courses of BCA program will provide a holistic approach to technical education, giving students an overview of the field, a basis to build and specialize upon practical learning. These laboratory courses are the strong foundation to establish practical knowledge and skills in the respective field of computer science. We offer laboratory courses in semester I, II, III, IV, V & VI, and VII during the BCA program. There will be 1 credits for each laboratory course and 3 credits each for theory classes.

II. Discipline Major Courses:

Discipline Major Courses cover a broad range of courses to equip students with a solid foundation in basic accounting principles, financial statements, and the analysis of a company's financial performance, product development, pricing strategies, promotion, and distribution of products and services. The Discipline Major Courses and Laboratory courses of BCA program will provide a holistic approach to technical education, giving students an overview of the field, a basis to build and specialize upon practical learning. These laboratory courses are the strong foundation to establish practical knowledge and skills in the respective field of computer science.

These major courses, Marketing, Human Resource Management, International Business, Finance, Digital Marketing and Business Analytics, Fintech and Block Chain Technology - provide a comprehensive understanding of the business world and prepare Students for various careers in management, marketing, finance, human resources, and entrepreneurship.

These major courses, Full Stack, Artificial intelligence, Data Analytics, Data Science, Cyber security, Computer Science Web Applications Cloud Computing and IOT Software Engineering & Quality Testing Digital Marketing Cloud Technology and Information Technology - provide a comprehensive understanding of the business world and prepare Students for various careers in computer science, Artificial Intelligence, Data science, Data Analytics.

III. Discipline Minor Courses:

Discipline Minor courses in a BCA program offer students the opportunity to explore additional areas outside their major, enhancing their knowledge and skill set in specific areas of interest. These courses complement the primary management /Computer application curriculum and provide a well-rounded education. Minor courses allow students to diversify their expertise and tailor their education to align with their career goals and interests. The combination of major and minor courses enriches their skill set, making Students more versatile and adaptable in the competitive job market/industry and enhances their ability to solve real life problems.

IV. Skill Enhancement Courses (SEC):

Skill Enhancement Courses (SECs) in a BCA program are designed to complement the core curriculum and provide students with additional skills and experiences that enhance their employability and professional development. These programs aim to bridge the gap between theoretical knowledge and practical application, making Students more competitive and well-rounded in the job market/ job industry. Skill Enhancement Courses not only add value to students' education but also equip them with a unique set of skills that make them more adaptable, innovative, and marketable in the dynamic business landscape. A Skill Enhancement Course is a compulsory course of 1-credit each course, which is basically meant to enhance general ability of students in areas like soft skills, quantitative aptitude and reasoning ability - required for the overall development of a student and at the same time crucial for industry/corporate demands and requirements.

V. Ability Enhancement Courses (AEC):

Ability Enhancement Courses (AECs) in a BCA program are designed to enhance students' abilities and skills beyond their core curriculum. AEC courses are the courses based upon the content that leads to knowledge enhancement through various areas of study. They are based on Language and Literature, and Environmental Science which are mandatory for all disciplines. These courses aim to provide a well-rounded education and equip students with essential competencies required to excel in their personal and professional lives. Ability Enhancement Courses contribute to students' overall growth and prepare them for the challenges of the modern business world. By developing a diverse skill set, students become better equipped to succeed in their careers and make a positive impact in the organizations they join. These courses also foster a culture of continuous learning and self-improvement, which is essential for professional development in the long run. Curriculum offers 4 courses of 2 credits each which include: English Communication I & II, Content Writing & Editing and Communication tools in digital era.

VI. Value Added Courses (VAC):

Value Added Courses (VACs) in a BCA program are additional courses that provide students with specialized

knowledge and skills beyond the regular curriculum. These courses are designed to add value to the students' education and make them more competitive and employable in the job market. Value Added Courses not only enhance students' knowledge and skills but also demonstrate their commitment to continuous learning and professional development. These courses play a crucial role in shaping well-rounded and versatile Students who are ready to tackle the challenges of the industry/corporate business world and make meaningful contributions to their organizations.

VII. Summer Internship Report :

Summer internships are an integral part of the BCA program, offering students valuable hands-on experience in a real-world environment during their summer break. It helps to provide the industrial exposure to the students. These internships typically last for a 6 to 8 weeks and allow students to apply the theoretical knowledge gained in their courses to practical scenarios. Overall, summer internships are a vital component of a BCA program, providing students with a practical understanding of the industrial world, valuable skills, and a head start in their professional journey. It is a mutually beneficial experience for both students and companies, as businesses gain fresh perspectives and potential future employees, while students gain real-world experience and exposure to their chosen career paths.

VIII. Major Project Report:

The project and Industrial training has to be taken as per the guideline issued from time to time. It helps to provide the industrial exposure to the students. They are being able to learn, enhance their skills and utilize the learnt concept to be able to understand the facts practically.

A Major Project Report is a comprehensive document that provides insights into Software requirements specification regarding, trends, and conditions.

All students pursuing BCA Shall submit the proposed topic of the seminar (Field Project) in the first week of the semester to the course coordinator. Care should be taken that the topic selected does not directly relate to the subject of the courses being pursued. The course coordinator shall then forward the list to the concerned Seminar Committee. The topics will then be allocated to the students along with the name of the faculty guide.

Preparation of the seminar

1. The student shall meet the guide for the necessary guidance for the seminar (field project) work.
2. During the next two to four weeks the student should read the primary literature germane to the seminar topic. Reading selection should continuously be informed to the guide.

After necessary collection of data and literature survey, the students must prepare a report.

IX. Industry Project Report:

An Industry Project Report (IPR) for a BCA program is a comprehensive document that reflects the practical application of theoretical knowledge gained during the course. It typically involves working on a real-world problem or project within an industry, providing students with valuable hands-on experience. 12 credits are being assigned to it .

X. Research Project Report:

The project and Industrial training has to be taken as per the guideline issued from time to time. It helps to provide the industrial exposure to the students. They are being able to learn, enhance their skills and utilize the learnt concept to be able to understand the facts practically. 20 credits are being assigned to it .

In a BCA program, the research project report holds significant importance as it serves multiple critical purposes:

1. Academic Requirement: A research project report is often a culmination of academic learning throughout the BCA program. It demonstrates a student's ability to integrate theoretical knowledge gained in courses with practical application in a real-world context.
2. Practical Application: Unlike theoretical exams or assignments, a research project report requires students to apply their knowledge to address a real business problem or explore a relevant research topic. This hands-on approach fosters critical thinking, problem-solving skills, and the ability to analyze data.
3. Skill Development: Writing a research project report enhances various skills essential for a professional.
 - a. Research Skills: Students learn to gather relevant information using both primary and secondary (literature review, reports) sources.
 - b. Analytical Skills: They analyze data collected to derive meaningful conclusions and recommendations.
 - c. Communication Skills: Writing the report improves their ability to communicate complex ideas clearly and concisely, catering to both academic and professional audiences.
4. In-depth Understanding: It encourages a deeper exploration of a specific business area of interest. By

conducting research, students gain insights that go beyond what is covered in standard coursework, thereby enriching their understanding of real world problem.

5. Professional Preparation: Completing a research project prepares students for the demands of the professional world. Businesses value employees who can conduct research, analyze findings, and present actionable recommendations—skills directly honed through the process of creating a research project report

XI. Learning by Project Based Entrepreneurial Initiative (LPBEI)

Learning by project-based entrepreneurial initiative refers to an educational approach where students engage in hands-on, practical projects with an entrepreneurial focus. Instead of relying solely on traditional classroom instruction, students work on real- world projects that simulate entrepreneurial experiences and challenges. These initiatives aim to provide students with opportunities to develop essential entrepreneurial skills, such as critical thinking, problem-solving, creativity, teamwork, and business acumen. The emphasis is not only on the outcome of the project but also on the learning process itself. Students are encouraged to actively explore and apply theoretical concepts, gather and analyze data, make informed decisions, and reflect on their experiences. Through these initiatives, students gain practical experience, learn to navigate real-world challenges, and develop the skills and mindset necessary for entrepreneurship. Total 24 credits are being assigned to it with 4 courses of 2 credit each in III, IV,V& VI Semester, 4 credit in VIIth semester, 1 course of 12 credit in VIII semester.

For BCA (H) 4 Years Program

S.No .	Types of Course	No. of Courses	Credit per Course	Total Credit
1	Discipline Core Courses	8	4	32
2	Discipline Major Courses	10+2 (MOOC)	4	48
3	Discipline Minor Courses	8	4	32
4	Interdisciplinary Minor Courses	4	4	16
5	Value Added Courses (VAC)	4	2	8
6	Ability-Enhancement Course (AEC)	8	2	16
7	Skill-Enhancement Elective Course (SEC)	4	3	12
8	PBL/ LPBEI-1	1	2	2
9	PBL/ LPBEI-2	1	2	2
10	PBL/ LPBEI-3	1	2	2
11	Summer Internship Report	1	2	2
12	Major Project Report	1	4	4
13	Industry Project Report/ Research Project Report	1	12/20	12
Total		54		188

XII. Program Specific Outcomes (PSOs): After completing the programme, students will be able to achieve the following specific outcomes:

PSO 1:		Understand how to identify, formulate and design solutions in the areas of Computer Science and Application.
PSO 2:		Demonstrate the abilities to design and develop algorithms and implement them as programs, with analysis and interpretation of data.
PSO 3:		Develop skills in software development so as to enable the graduates to take up employment/ self-employment in local, Indian & global software market.
PSO 4:		Enhance Employability by developing leadership, effective communication & time management skills and also by incorporating ethics & team work ability.

D. Program Outcome (POs): The program outcome for a BCA degree may vary depending on the institution offering the program and its particular focus. POs should be measurable and directly related to the program's curriculum and outcomes.

PO 1:	Critical thinking:	Take informed actions after identifying the assumptions that frame our thinking and actions, checking out the degree to which these assumptions are accurate and valid, and looking at our ideas and decisions (intellectual, organizational, and personal) from different perspectives.
PO 2:	Effective communication :	Speak,read,write and listen clearly in person and through electronic media in English and in one Indian language, and make meaning of the world by connecting people, ideas, books ,media and technology
PO 3:	Social interaction :	Social interactions also play important role to reads the conclusion in group settings.
PO 4:	Effective citizenship:	Demonstrate Empathetic social concern and equity centered national development, and the ability to act with an informed awareness of issues and participate in civic life through volunteering.
PO 5:	Ethics:	It has direct impact to recognize the different value systems. It gives proper understanding in different dimension form a king decisions.
PO 6:	Environment and sustainability	Recognize different value systems including your own, understand the moral dimensions of your decisions, and accept responsibility for them.
PO 7:	Self-directed & lifelong learning:	Acquire the ability to engage in independent and life-long learning in broad spectrum including socio technological changes.
PO 8:	Problem analysis & Solving:	Identify, formulate, research literature, and analyze complex basic sciences problems reaching substantiated conclusions using first principles of mathematics, natural sciences.
PO 9:	Entrepreneurship:	An Entrepreneurship cut a cross every sector of human life including the field of engineering, engineering entrepreneurship is the process of harnessing the business opportunities in engineering and turning it into profitable commercially viable innovation.
PO 10:	Interpersonal skills Learning:	Inter personal skills involve the ability to communicate and build relationships with others. Effective interpersonal skills can help the students during the job interview process and can have a positive impact on your career advancement.
PO 11:	Technology savvy/usage:	Being technology savvy is essentially one's skill to be smart with technology. This skill reaches far beyond 'understanding' the concepts of how technology works and encompasses the'utilization'of such modern technology for the purpose of enhancing productivity and efficiency.

E. Pedagogy & Unique practices adopted:

“Pedagogy is the method and practice of teaching, especially for teaching an academic subject or theoretical concept”. In addition to conventional time-tested lecture method, the institute will emphasize on experiential learning:

1. Case Based Learning: Case based learning enhances student skills at delineating the critical decision dilemmas faced by organizations, helps in applying concepts, principles and analytical skills to solve the delineated problems and develops effective templates for technical problem solving. Case method of teaching is used as a critical learning tool for effective learning and we encourage it to the fullest. We make it compulsory to teach at least one case study in each unit of every course in BCA program.

2.

3. Role Play & Simulation: Role-play and simulation are forms of experiential learning. Learners take on different roles, assuming a profile of a character or personality, and interact and participate in diverse and complex learning settings. Role-play and simulation function as learning tools for teams and groups or individuals as they "play" online or face-to-face. They alter the power ratios in teaching and learning relationships between students and educators, as students learn through their explorations and the viewpoints of the character or personality they are articulating in the environment. This student-centered space can enable learner- oriented assessment, where the

design of the task is created for active student learning. Therefore, role-play & simulation exercises such as virtual share trading, technical simulation etc. are being promoted for the practical-based experiential learning of our students.

4. Video Based Learning (VBL) & Learning through Movies (LTM): These days technology has taken a front seat and classrooms are well equipped with equipment and gadgets. Video-based learning has become an indispensable part of learning. Similarly, students can learn various concepts through movies. In fact, many teachers give examples from movies during their discourses. Making students learn few important theoretical concepts through VBL & LTM is a good idea and method. The learning becomes really interesting and easy as videos add life to concepts and make the learning engaging and effective. Therefore, our institute is promoting VBL & LTM, wherever possible.

5. Field/Live Projects: The students, who take up experiential projects in companies, where senior executives with a stake in teaching guide them, drive the learning. All students are encouraged to do some live project other their regular classes.

6. Industrial Visits: Industrial visit are essential to give students hand-on exposure and experience of how things and processes work in industries. Our institute organizes such visits to enhance students' exposure to practical learning and work out for a report of such a visit relating to their specific topic, course or even domain.

7. MOOCS: Students may earn credits by passing MOOCS as decided by the college from time to time. Graduate level programs may award Honors degree provided students earn pre-requisite credits through MOOCs.

8. Special Guest Lectures (SGL) & Extra Mural Lectures (EML): Some topics/concepts need extra attention and efforts as they either may be high in difficulty level or requires experts from specific industry/domain to make things/concepts clear for a better understanding from the perspective of the industry. Hence, to cater to the present needs of industry we organize such lectures, as part of lecture-series and invite prominent personalities from academia and industry from time to time to deliver their vital inputs and insights.

9. Student Development Programs (SDP): Harnessing and developing the right talent for the right industry an overall development of a student is required. Apart from the curriculum teaching various student development programs (training programs) relating to soft skills, interview skills, SAP, Advanced ms-office training etc. that may be required as per the need of the student and industry trends, are conducted across the whole program. Participation in such programs is solicited through volunteering and consensus

10. Industry Focused programme: Establishing collaborations with various industry partners to deliver the programme on sharing basis. The specific courses are to be delivered by industry experts to provide practice based insight to the students.

11. Orientation Program: Student orientation program plays an important role in a student's transition to a university life. Orientation programs are aimed at familiarizing the students to an unknown campus environment, its faculties and infrastructure. It enables them to make essential connection with studies and develop network among other peers.

The orientation program would be of approximately 2 weeks. The main purpose of the orientation program is to make the students aware of institute policy rules and assets so that students would feel comfortable when they join that college. Various topics would be covered in the orientation program like:

- Introducing students to their college life.
- Incorporating them in the university environment.
- Giving the opportunity to the university members and faculties to get connected to the new batch.
- Awareness of linkages among Society, Environment, Education & Development.
- Philosophy of Education, Indian Education System and Pedagogy.
- Background for enriching Subject Knowledge.

- Information Technology and Computer Awareness.

12. Mentoring Scheme: Mentoring schemes aim to provide a forum in which students can talk about common issues such as their career development, share their experiences and discuss challenges they are facing, their problems related with college or outside college. They talk with their mentors and mentors provide them solutions.

13. Special Assistance Program for slow learners & fast learners: write the note how would you identify slow learners, develop the mechanism to correcting knowledge gap. Terms of advance topics what learning challenging, it will be provided to the fast learners.

14. Career & Personal Counseling: It is a process that will help the student to know and understand yourself and the world of work in order to make career, educational, and life decisions. Career development is more than just deciding on a major and what job you want to get when you graduate. Corporate Resource Cell or Placement Cell provides support to shape the students to a brighter future. The CRC majorly helps students by:

➤ **Career Counseling:**

The main reason of consulting this department was your uncertainty about career prospects. With the view of catering to this need of the students, the Corporate Resource Centers have employed proficient counselors who are familiar with the contemporary industry trends. CRCs also provide guidance and mentoring for all choice based career options like Placements, Entrepreneurship, Higher Studies services. By knocking their door, you would be able to bring your thoughts to the table and in return, they would be able to guide you in the right direction.

➤ **Workshops:**

Before expanding your skills and way before propelling you in the industry, the department will brief you about the requirements through regular workshops. Through these workshops, the students are given insights on the topics such as designing an impressive CV with the assistance of latest tools and techniques, the use of different applications, etc.

➤ **All-inclusive Training:**

Placement centers have a full hold on the training activities. In furtherance of sharpening the skills apropos to recent demands, these centers conduct interwoven training sessions like Training on Aptitude, Logical Reasoning and Quantitative skills. Employability Assessment Test is conducted periodically to check the progress of training and assess the skill level of the student for various sectors of jobs. Also training is provided on latest technologies which are relevant to the industry needs of today. From communication skills to clothing etiquette, these centers incorporate each and every aspect of your personality. This training is the tipping point which instills full confidence and desired skills in you.

➤ **Mock Interviews:**

Interviews, for some, are herculean tasks while for others they are child's play. Undoubtedly, it is all intrinsic to our nature but these things can be transfigured with some tactics. Now, those techniques which construct a smooth road to success are with career services center workers-they know what needs to be done. So, with the help of mock interviews, your nervousness is reduced and one gets an opportunity to improve on the weak areas. This indeed proves to be a great help for the final face-off!

➤ **Internships and Industrial Visits:**

For unveiling the real-life working and functions of any company, short industrial trips become an integral part of the curriculum. These are arranged and organized by the same department. Internships are another alternative to providing complete know-how of the company. Here, the students undergo training at the actual physical locations. These are of utter importance as they give the students the much required experiential learning. These internship programs let the students ponder their choices. Through these visits, the students are exposed to situations and more adapt to taking on the challenges as they venture into the professional workspace.

➤ **On / Off Campus placement:**

Placement centers are the driving forces behind on-campus recruitments. Those who work under the head of this department make a strategic alliance with a gamut of companies. During and towards the end of the session, companies visit the campus. As inexperienced candidate may face trouble finding a befitting job, so these centers actually help them in bypassing that trouble and become more confident and prepared.

All in all, consulting your placement center is always a great option for it resolves numerous career-related issues. Taking its working and functioning into account, placement center emerges as a helpful department. From counseling the students to boosting morale, it provides every possible help needed. While adding more to your portfolio, it gives back-end support for a better career avenue to the student and in regard to this, one should never ignore it.

15. Competitive Exam Preparation: Competitive exams require a certain mindset and understanding which is quite different from a regular school or college academic test. Aptitude, Logical Reasoning, Computer Questions will help the students to prepare for Online Exam. Apart from revision, another important thing is to practice. This practice

can include previous year's paper and the test papers of the exams that are available online.

16. Extracurricular Activities: An extracurricular activity is an activity that is not required by the course of study at your university. They are not obligatory but are invaluable in developing your talents and practical skills. Some of these activities, such as fundraising and volunteering, not only benefit you but also help others. Organizing & participation in extracurricular activities will be mandatory to help students develop confidence & face audience with care.

17. Participation in Seminars/ Workshop and Writing and Presenting Papers: As per the requirement of course, students have to participate in various seminars/ workshops. Students have to develop writing research articles and also develop the presentation skills under the guidance of faculty members.

18. Library Visit & Utilization of E-Learning Resources: Library is the physical building or a room with the collection of books and resources accessible to a community. It consists of books and e-learning resources on diverse genres and subjects. Libraries are important for healthy development of society. Libraries provide valuable services to meet the learning needs of the people. Libraries also benefit the economy of our nation as people use them for research purposes and to improve their job skills. They play an essential part in overall educational development of people and community. Student can visit the library from morning 10 AM to evening 8 PM. Library created its resources Database and provided Online Public Access Catalogue (OPAC) through which users can be accessed from any of the computer connected in the LAN can know the status of the book. Now we are in process to move from OPAC to KOHA.

- a) Institute Library & Information is subscribing online e-books and e-journals databases (DELNET and EBSCO host E-databases) as per the requirement of the institute and fulfilling AICTE norms. IP based access is given to all computers connected on campus LAN to access e-journals.
- b) For the effective utilization of resources, Information Literacy training programs are conducted to the staff and students.
- c) Wi-Fi enabled campus
- d) Regular addition of latest books and journals
- e) Well maintained e-library to access e-resources

19. Capability Enhancement & Development Schemes: Time to time program coordinator shall organize capability enhancement program for students which are essential to complete the degree which are not covered in the syllabus.

Study & Evaluation Scheme

Bachelor of Computer Applications (B.C.A.)

Semester -1

S.No	Category	Course Code	Course Name	L	T	P	Credits	Internal	External	Total
1	Major	MJC-1	One course from already been selected as Major specialization in first semester. AI/DA/CS/FS/W A/ST/CI/DM/CTI S/MAWT	4	0	0	4	40	60	100
2	Core	BCACC240101	Computer Fundamentals & Programming concepts	3	0	0	3	40	60	100
3	Minor	MNC-1	One course from already been selected as Minor specialization in first semester. AI/DA/CS/FS/W A/ST/CI/DM/CTI	4	0	0	4	40	60	100

			S/MAWT							
4	AEC	TMUAE240101	English Communication -I	1	0	2	2	40	60	100
5	VAC	TMUVA240101	Indian Knowledge System	2	0	0	2	40	60	100
6	Inter Disciplinary	IDC-1	Inter – Disciplinary Course-I*	4	0	0	4	40	60	100
7	SEC	TMUSE240401	Modern office Management	3	0	0	3	40	60	100
8	Core	BCACC240151	Computer Fundamentals & Programming concepts (Lab)	0	0	2	1	50	50	100
Total				21	0	4	23	330	470	800
Semester -II										
S.No .	Category	Course Code	Course Name	L	T	P	Credits	Internal	External	Total
1	Major	MJC-2	One course from already been selected as Major specialization in first semester. AI/DA/CS/FS/WA/ST/CI/D M/CTIS/MAWT	4	0	0	4	40	60	100
2	Core	BCACC240201	Data Structures Using C++	3	0	0	3	40	60	100
3	Minor	MNC-2	One course from already been selected as Minor specialization in first semester. AI/DA/CS/FS/WA/ST/CI/DM/CTIS/MAWT	4	0	0	4	40	60	100
4	SEC	TMUSE240101	Innovation & Entrepreneurship Development	3	0	0	3	40	60	100
5	VAC	TMUVA240201	Environmental Studies & Disaster Management	2	0	0	2	40	60	100
6	Inter-Disciplinary	IDC-II	Inter-Disciplinary Course– II**	4	0	0	4	40	60	100
7	Core	BCACC240251	Data	0	0	2	1	50	50	100

			Structures Using C++ Lab							
Total				20	0	2	21	290	410	700
Discipline Core Courses										
S.No .	Category	Course Code	Course Name	L	T	P	Credits	Internal	External	Total
1	First	BCACC240101	Computer Fundamentals and C Programming	3	0	2	4	40	60	100
2	Second	BCACC240201	Data Structures Using C	3	0	2	4	40	60	100
3	Third	BCACC240301	Operating System	3	0	2	4	40	60	100
4	Fourth	BCACC240401	Digital Electronics and Computer Organization	3	1	0	4	40	60	100
5	Fifth	BCACC240501	Software Engineering and Testing	3	1	0	4	40	60	100
6	Sixth	BCACC240601	Computer Networks	3	0	2	4	40	60	100
7	Seventh	TMUCC240701	Research Methodology	3	0	2	4	40	60	100
8	Seventh	TMUCC240702	Project Management	3	0	2	4	40	60	100
Discipline Major/Minor Courses										
(Specialization: Artificial Intelligence (AI), Data Analytics (DA), Cyber security (CS), Full Stack (FS), Web Applications (WA), Software Engineering & Quality Testing (ST), Cloud Computing and IOT (CI), Digital Marketing (DM), Cloud Technology and Information Security (CTIS) and Mobile Application and Web Technology (MAWT))										
Artificial Intelligence (AI)										
S.No .	Semester	Course Code	Course Name	L	T	P	Credits	Internal	External	Total
1	First	BCAAI240101	Artificial Intelligence	3	0	2	4	40	60	100
2	Second	BCAAI240301	Introduction to Machine Learning	3	0	2	4	40	60	100
3	Third	BCAAI240201	Python programming	3	0	2	4	40	60	100
4	Fourth	BCAAI240401	Basic of statistics	3	0	2	4	40	60	100
5	Fifth	BCAAI240501	Artificial Neural Network	3	0	2	4	40	60	100
6	Fifth	BCAAI240502	Fundamental of Pandas ,Skleran , seaborn & matplotlib libraries	3	0	2	4	40	60	100
7	Sixth	BCAAI240601	Introduction to Natural language processing(NLP)	3	0	2	4	40	60	100
8	Sixth	BCAAI240602	Fundamental of Machine Learning Models Evaluation Parameters	3	0	2	4	40	60	100

9	Seventh	BCAAI240701	Deep learning	3	0	2	4	40	60	100
10	Seventh	BCAAI240702	Genetic Algorithms	3	0	2	4	40	60	100
Data Analytics (DA)										
S.No .	Category	Course Code	Course Name	L	T	P	Credits	Internal	External	Total
1	First	BCADA24 0101	Fundamental of Data analytics Techniques	3	0	2	4	40	60	100
2	Second	BCADA24 0201	Probability & Statistics	3	0	2	4	40	60	100
3	Third	BCADA24 0301	Data Warehouse and Mining	3	0	2	4	40	60	100
4	Fourth	BCADA24 0401	Data Optimization Techniques	3	0	2	4	40	60	100
5	Fifth	BCADA24 0501	Introduction to Advanced Excel used in Data analytics	3	0	2	4	40	60	100
6	Fifth	BCADA24 0502	Introduction to Tableau	3	0	2	4	40	60	100
7	Sixth	BCADA24 0601	Introduction to Power BI	3	0	2	4	40	60	100
8	Sixth	BCADA24 0602	R-Programming	3	0	2	4	40	60	100
9	Seventh	BCADA24 0701	Introduction to Apache Spark	3	0	2	4	40	60	100
10	Seventh	BCADA24 0702	Julia Programming	3	0	2	4	40	60	100
Cyber security (CS)										
S.No .	Category	Course Code	Course Name	L	T	P	Credits	Internal	External	Total
1	First	BCACS24 0101	Fundamentals of Cyber Security	3	0	2	4	40	60	100
2	Second	BCACS24 0201	Concept of Ethical hacking	3	0	2	4	40	60	100
3	Third	BCACS24 0301	Cyber-crime and cyber law	3	0	2	4	40	60	100
4	Fourth	BCACS24 0401	Social Media Overview and security	3	0	2	4	40	60	100
5	Fifth	BCACS24 0501	E commerce and digital payments	3	0	2	4	40	60	100
6	Fifth	BCACS24 0502	Networking security	3	0	2	4	40	60	100
7	Sixth	BCACS24 0601	Cyber Security Management, Compliance and Governance	3	0	2	4	40	60	100
8	Sixth	BCACS24 0602	Digital Devices Security, tools, and Technologies for Cyber Security	3	0	2	4	40	60	100
9	Seventh	BCACS24 0701	Cyber forensic investigation	3	0	2	4	40	60	100
10	Seventh	BCACS24	Advanced Cyber	3	0	2	4	40	60	100

		0702	Security							
Full Stack (FS)										
S.No .	Category	Course Code	Course Name	L	T	P	Credits	Internal	External	Total
1	First	BCAFS24 0101	JAVA	3	0	2	4	40	60	100
2	Second	BCAFS24 0201	Agile	3	0	2	4	40	60	100
3	Third	BCAFS24 0301	Advanced HTML CSS, Java Script	3	0	2	4	40	60	100
4	Fourth	BCAFS24 0401	Introduction to React Basics and building Components	3	0	2	4	40	60	100
5	Fifth	BCAFS24 0501	Introduction to SQL	3	0	2	4	40	60	100
6	Fifth	BCAFS24 0502	Mango DB	3	0	2	4	40	60	100
7	Sixth	BCAFS24 0601	Node JS	3	0	2	4	40	60	100
8	Sixth	BCAFS24 0602	Express JS	3	0	2	4	40	60	100
9	Seventh	BCAFS24 0701	DAA	3	0	2	4	40	60	100
10	Seventh	BCAFS24 0702	Introduction to Restful API	3	0	2	4	40	60	100
Web Applications (WA)										
S.No .	Category	Course Code	Course Name	L	T	P	Credits	Internal	External	Total
1	First	BCAWA2 40101	Django	3	0	2	4	40	60	100
2	Second	BCAWA2 40201	PHP with My SQL	3	0	2	4	40	60	100
3	Third	BCAWA2 40301	Introduction to Servlet	3	0	2	4	40	60	100
4	Fourth	BCAWA2 40401	Introduction to JSP	3	0	2	4	40	60	100
5	Fifth	BCAWA2 40501	Web technologies	3	0	2	4	40	60	100
6	Fifth	BCAWA2 40502	Node JS	3	0	2	4	40	60	100
7	Sixth	BCAWA2 40601	Fundamental of Angular JS	3	0	2	4	40	60	100
8	Sixth	BCAWA2 40602	Fundamental of Web Designing	3	0	2	4	40	60	100
9	Seventh	BCAWA2 40701	ASP .Net with C#	3	0	2	4	40	60	100
10	Seventh	BCAWA2 40702	Fundamental of XML	3	0	2	4	40	60	100
Software Engineering & Quality Testing (ST)										
S.No	Category	Course	Course Name	L	T	P	Credits	Internal	External	Total

.		Code								
1	First	BCAST24 0101	Introduction to Selenium	3	0	2	4	40	60	100
2	Second	BCAST24 0201	Software Testing: Principles, Techniques and Tools	3	0	2	4	40	60	100
3	Third	BCAST24 0301	Fundamental of Search Engine Optimization(SEO)	3	0	2	4	40	60	100
4	Fourth	BCAST24 0401	Fundamental of software testing models	3	0	2	4	40	60	100
5	Fifth	BCAST24 0501	Software Requirement Analysis and Specification	3	0	2	4	40	60	100
6	Fifth	BCAST24 0502	Software Project Management	3	0	2	4	40	60	100
7	Sixth	BCAST24 0601	Fundamental of Automated Software Testing	3	0	2	4	40	60	100
8	Sixth	BCAST24 0602	Software Verification and Validation	3	0	2	4	40	60	100
9	Seventh	BCAST24 0701	Introduction to UML	3	0	2	4	40	60	100
10	Seventh	BCAST24 0702	Software Maintenance and Configuration	3	0	2	4	40	60	100
Cloud Computing and IOT (CI)										
S.No .	Semester	Course Code	Course Name	L	T	P	Credits	Internal	External	Total
1	First	BCACI24 0101	Introduction To Cloud Computing	3	0	2	4	40	60	100
2	Second	BCACI24 0201	Introduction to Amazon Web Services (AWS)	3	0	2	4	40	60	100
3	Third	BCACI24 0301	Power BI	3	0	2	4	40	60	100
4	Fourth	BCACI24 0401	Introduction to Salesforce	3	0	2	4	40	60	100
5	Fifth	BCACI24 0501	Virtualization and Cloud Security	3	0	2	4	40	60	100
6	Fifth	BCACI24 0502	Introduction to Big Data	3	0	2	4	40	60	100
7	Sixth	BCACI24 0601	IOT and its Application	3	0	2	4	40	60	100
8	Sixth	BCACI24 0602	Cloud Web Services	3	0	2	4	40	60	100
9	Seventh	BCACI24 0701	Introduction to Azure	3	0	2	4	40	60	100
10	Seventh	BCACI24 0702	Fundamental of Hadoop	3	0	2	4	40	60	100
Digital Marketing (DM)										
S.No .	Semester	Course Code	Course Name	L	T	P	Credits	Internal	External	Total
1	First	BCADM2 40101	Basics of Digital Marketing	3	0	2	4	40	60	100
2	Second	BCADM2 40201	Web Fundamentals	3	0	2	4	40	60	100

3	Third	BCADM2 40301	Social Media Marketing	3	0	2	4	40	60	100
4	Fourth	BCADM2 40401	E-commerce	3	0	2	4	40	60	100
5	Fifth	BCADM2 40501	Search engine optimization	3	0	2	4	40	60	100
6	Fifth	BCADM2 40502	Audio & Video Editing	3	0	2	4	40	60	100
7	Sixth	BCADM2 40601	Web and Google Analytics	3	0	2	4	40	60	100
8	Sixth	BCADM2 40602	Cyber Laws and Digital Media Laws	3	0	2	4	40	60	100
9	Seventh	BCADM2 40701	Email and Mobile Marketing	3	0	2	4	40	60	100
10	Seventh	BCADM2 40702	Brand Management	3	0	2	4	40	60	100
Cloud Technology and Information Security (CTIS)										
S.No .	Semester	Course Code	Course Name	L	T	P	Credits	Internal	External	Total
1	First	BCACT2 40101	Introduction to Cloud Technology	3	0	2	4	40	60	100
2	Second	BCACT2 40201	Linux Administration	3	0	2	4	40	60	100
3	Third	BCACT2 40301	Principles of Virtualization	3	0	2	4	40	60	100
4	Fourth	BCACT2 40401	Cyber Security Fundamentals	3	0	2	4	40	60	100
5	Fifth	BCACT2 40501	Ethical Hacking	3	0	2	4	40	60	100
6	Fifth	BCACT2 40502	Cloud Web Services	3	0	2	4	40	60	100
7	Sixth	BCACT2 40601	Digital Forensics and Investigation	3	0	2	4	40	60	100
8	Sixth	BCACT2 40602	Cloud Migration	3	0	2	4	40	60	100
9	Seventh	BCACT2 40701	Malware Analysis	3	0	2	4	40	60	100
10	Seventh	BCACT2 40702	Automation and Configuration Management	3	0	2	4	40	60	100
Mobile Application and Web Technology (MAWT)										
S.No .	Semester	Course Code	Course Name	L	T	P	Credits	Internal	External	Total
1	First	BCAMW 240101	Introduction to Mobile Device and Network Architecture	3	0	2	4	40	60	100
2	Second	BCAMW 240201	User Interface and User Experience	3	0	2	4	40	60	100
3	Third	BCAMW 240301	PHP Programming	3	0	2	4	40	60	100
4	Fourth	BCAMW 240401	Android Application Development	3	0	2	4	40	60	100
5	Fifth	BCAMW 240501	Web 3.0	3	0	2	4	40	60	100
6	Fifth	BCAMW	Advanced Android	3	0	2	4	40	60	100

		240502	Application Development							
7	Sixth	BCAMW 240601	JavaScript Frameworks	3	0	2	4	40	60	100
8	Sixth	BCAMW 240602	Version Control Systems	3	0	2	4	40	60	100
9	Seventh	BCAMW 240701	Web Development using Python	3	0	2	4	40	60	100
10	Seventh	BCAMW 240702	iOS Application Development	3	0	2	4	40	60	100
Value Added Courses										
S.No	Category	Course Code	Course Name	L	T	P	Credits	Internal	External	Total
1	First	TMUVA 240101	Indian Knowledge System	1	0	2	2	40	60	100
2	Second	TMUVA 240201	Environmental Studies & Disaster Management	1	0	2	2	40	60	100
3	Third	TMUVA 240301	Digital Technology in Education	1	0	2	2	40	60	100
4	Fourth	TMUVA 230401	Human Values & Professional Ethics	2	0	0	2	40	60	100
Ability Enhancement Courses										
S.No	Category	Course Code	Course Name	L	T	P	Credits	Internal	External	Total
1	First	TMUAE 240151	English Communication	1	0	2	2	50	50	100
2	Fourth	TMUTP 240401	Object-Oriented Programming Concepts Using C++	1	0	2	2	40	60	100
3	Fifth	TMUAE 240551	Personal Effectiveness	2	0	0	2	40	60	100
4	Fifth	TMUTP 240501	Database Management Systems	0	0	4	2	50	50	100
5	Sixth	TMUTP 240601	Object-Oriented Programming Concepts Using Java	2	0	0	2	40	60	100
6	Sixth	TMUAE 240601	Team Building and Leadership	2	0	0	2	40	60	100
7	Seventh	TMUAE 240701	Corporate Skills	1	0	2	2	40	60	100
8	Seventh	TMUTP 240701	Web Programming	1	0	2	2	40	60	100
Skills Enhancement Courses										
S.No	Category	Course Code	Course Name	L	T	P	Credits	Internal	External	Total
1	First	TMUSE 240101	Innovation & Entrepreneurship Development	2	0	2	3	40	60	100
2	Second	TMUSE2 40201	Basic Analytical Skills	2	0	2	3	40	60	100
3	Third	TMUSE2 40301	Advanced Analytical Skill	2	0	2	3	40	60	100
4	Fourth	TMUSE 240401	Modern office Management	2	1	0	3	40	60	100

Project Based Learning										
S.No	Category	Course Code	Course Name	L	T	P	Credits	Internal	External	Total
1	Third	BCAPL240301	PBL-1	1	0	2	2	50	50	100
2	Fourth	BCAPL240401	PBL-2	1	0	2	2	50	50	100
3	Fifth	BCAPL240501	PBL-3	0	0	4	2	50	50	100
4	Sixth	BCAPL240601	PBL-4	1	0	2	2	50	50	100
5	Seventh	BCAPL240701	Major Project	4	0	0	4	50	50	100
6	Eighth	BCAIP240801	Industry Project/Apprenticeship	6	0	12	12	50	50	100
Or (If Student opting for BCA Honors with Research)										
7	Eighth	BCARP240801	Research Project based on topic given by the Supervisor (Outcome one Research Paper to be published in UGC CARE/ SCOPUS/ SCI)	6	0	28	20	50	50	100
Note:- C-Credits 1C = 1 Hour of Lecture /Tutorial 1C = 2 Hour of Practical										
Detailed Syllabus										

Discipline Core Courses		
Course Code: BCACC240101	Discipline Core Course -1 BCA- Semester-I Computer Fundamentals & Programming concepts	L-3 T-0 P-0 C-3
Course Outcomes:	On completion of the course, the students will be:	
CO1.	Understanding the concept of various components of computer system	
CO2.	Understanding the basic programming Language constructs.	
CO3.	Analyzing basic mathematical problem and their solutions through programming	
CO4.	Applying knowledge to prepare programming solutions for distinct problems.	
CO5.	Applying knowledge to prepare scalable solutions through functions.	
Course Content		
Unit-1:	Concepts in Computer Application: Definition of Electronic Computer; History; Generations; Characteristics and Application of Computers; Classification of Computers; Functional Component of Computer: CPU, I/O devices, Type of Memory & Memory Hierarchy. Data and data types: Definitions, data, data types: Integer, Character, Float, String, etc. Constants and Variable Declaration; Token; Keywords; Identifier	8 Hours
Unit-2	Programming Language Classification & Computer Languages: Generation of Languages; Introduction to 4GLs; Translators; Assemblers; Compilers; Interpreters. Operators and Expressions: Numeric and relation operators; logical operator; bit operator: operator precedence and associatively .Conditionals (The if	8 Hours

	statement, The switch statement, ,Nested if else statement.	
Unit-3	Control Structure: Looping and Iteration (The for statement, The while statement, The do-while statement, break, continue, goto statements). Functions: (Function Prototyping, passing parameters (Actual and formal arguments), returning values, recursion, Call by Value; Call by Reference, Scope of variable (local/global), Storage classes: auto, extern, static, register, Library Function.	8 Hours
Unit-4	Introduction to Internet: Web Browser, Search Engine, Creating E-Mail account, Attaching documents, Sending and Receiving E-Mails. Arrays: Notation and representation; Manipulation of array elements; Multidimensional arrays.	8 Hours
Unit-5	Functions & Strings: Definition; Declaration; Call by Value; Call by Reference; Returns values and their types; Function calls. Structure & Dynamic Memory Allocation: Structures and union, run time memory allocation functions, Introduction of preprocessor directives.	8 Hours
Textbooks:	1. Sinha P. K., <i>Computer Fundamentals</i> , BPB Publications.	
Reference Books:	1. Peter Nortans, <i>Introduction to Computers</i> , Tata McGraw Hill. 2. YashavantKanetkar, <i>Let us C</i> , BPB Publications. 3. Leon & Leon, <i>Fundamental of Information Technology</i> , Vikas Publishing. 4. Kanter, <i>Managing Information System</i> , Prentice-Hall. <i>*Latest editions of all the suggested books are recommended.</i>	
Course Code: BCACC240151	Compulsory Core Course LAB -1 BCA Semester-I Computer Fundamentals &Programming concepts Part B	L-0 T-0 P-2 C-1
Course Outcomes:	On completion of the course, the students will be:	
CO1.	Applying programming concepts of functions and arrays to write compile and debug programs in C language.	
CO2.	Applying programming concepts of pointers and strings to write compile and debug programs in C language.	
CO3.	Applying programming concepts of structure and union to write compile and debug programs in C language.	
CO4.	Understanding a concept of object thinking within the framework of functional model	
CO5.	Understanding organization of the code to build large projects.	
Course Content:		
	LIST OF EXPERIMENTS- 1. To write a program to calculate Sum & average of N numbers. 2. To write a program to convert integer arithmetic to a given number of day and month. 3. To write a program to find maximum and minimum out of 3 numbers a, b & c. 4. To write a program to find factorial of positive integer. 5. To write a program to find sum of series up to n number, $\beta+5+8+\dots +n$. 6. To write a program to print all the number between 1 to 100 which are dividing by 7. 7. To write a program to generate Fibonacci series up to n. 8. To write a program to implement a function to calculate area of a circle. 9. To write a program to implement a recursive function to calculate factorial of given number.	

	10. To write a program to find whether number is prime or not. 11. To write a program to find that the enter character is a letter or digit. 12. To write a program to find addition of two matrix of n*n order. 13. To write a program to find multiplication of two matrix of n*n order. 14. To write a program to add 6 digit numbers in even case & multiple 6 digit number in odd case. 15. To write a program to find even or odd up to a given limit n. 16. To write a program to find whether a given no is palindrome or not. 17. To write a program to joining & Comparing the 2 string.	
Course Code: BCACC240201	Discipline Core Course -2 BCA Semester -II DATA STRUCTURES USING C++	L-3 T-0 P-0 C-3
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understanding and remembering the basic terminologies, compute the time Complexities and analyze the usage of array.	
CO-2	Analyzing the concept of Data Structures through ADT like stack, queue and linked list, also understand the basic usage and operations to be performed on them.	
CO-3	Understanding, analyzing and applying the learnt concept to solve problems related to various sorting and searching algorithm, later performing a comparative analysis of each one of them.	
CO-4	Understanding various representations of trees and graphs and analyzing different methods to solve various problems related with them.	
CO-5	Applying algorithm for solving problems like sorting, searching, insertion and deletion of data.	
Course Content		
Unit-1	Introduction: Basic Terminology, Elementary Data Organization, Data Structure operations, Algorithm, Complexity and Time-Space trade-off. Arrays: Array Definition, Representation and analysis, Single and Multidimensional Arrays, address calculation, application of arrays, Character String in C++, Character string operation.	8 Hours
Unit-2	Stack: Array Representation and Implementation of stack, Operations on Stack: Push & Pop, Linked Representation of Stack and Operations Associated with Stack, Applications of stack: Conversion of Infix to Prefix and Postfix Expressions, Evaluation of postfix expression using Stack. Queue: Array and linked representation and implementation of queues, Operations on Queue: Create, Add, Delete, and Circular queue.	8 Hours
Unit-3	Linked list: Representation and Implementation of Singly Linked List, Traversing and Searching of linked List, Overflow and Underflow, Insertion and deletion to/from Linked List, Insertion and deletion algorithms, Doubly linked list, Circular List, Linked List v/s Array.	8 Hours
Unit-4	Sorting: Bubble Sort, Selection Sort, Insertion Sort, Quick Sort, Merge Sort, and Heap Sort. Comparative Analysis of above Sorting algorithms. Searching: Sequential search, Binary Search.	8 Hours
Unit-5	Trees: Basic terminology, Binary Trees, Binary tree representation, algebaric Expressions, Complete Binary Tree, Array and Linked Representation of Binary trees, Traversing Binary tree, Binary Search Tree. Graph: Basic terminology, Graph representation using adjacency matrix, Graph representation using adjacency list.	8 Hours
Textbooks:	1. <i>Data Structures and Algorithm Analysis in C++</i> , Mark Allen Weiss 2. <i>Data Structures and Algorithms in C++</i> , Michael T. Goodrich, Roberto Tamassia, David Mount	
Reference Books:	1. <i>Object-Oriented Data Structures Using C++</i> , Nell Dale, Daniel T. Joyce, Chip Weems	

	2. <i>Data Structures Using C++, D. S. Malik</i> <i>* Latest editions of all the suggested books are recommended.</i>	
Course code: BCACC240251	Compulsory Core Course LAB -2 BCA Semester-I DATA STRUCTURE USING C++ LAB	L-0 T-0 P-2 C-1
CO -1	Applying the concepts for evaluating the operations on arrays , stack ,queue And linked list.	
CO-2	Analyzing and applying the techniques for solving problems related to searching And sorting.	
CO-3	Understanding the use of array representation.	
CO-4	Understanding the use of Binary Search Tree and applying the concept to Evaluate the operations to be performed on it.	
CO-5	Understanding and evaluating the time complexities of various algorithms and Data structure implemented for solving the problems.	
Course Content:		
	Array: Insertion of element in an array, deletion of element from an array.	
	Sorting: Selection Sort , Bubble Sort ,Insertion Sort, Quick Sort, Two Way Merge Sort and Heap Sort.	
	Searching: Sequential search, binary search.	
	Stack: Array Representation and Implementation of stack, Operations on Stacks: Push & Pop Conversion of Infix to Prefix and Postfix Expressions.	
	Queue: Array and linked representation and implementation of queues, Operations on Queue: Create Add, Delete, and Circular queue.	
	Linked list: Representation and Implementation of Singly Linked List, Traversing and Searching , Inserting and Deleting of Linked List. Same operation in Doubly Linked List, Circular Linked List.	
	Binary Search Tree: Creation, searching and traversal.	
Text Books:	1. <i>Lipschutz, DataStructure,TataMcGrawHill.</i>	
Reference Books:	1. <i>Kruseet.alR.,DataStructuresandProgramDesigninC,Pearson Education.</i> 2. <i>Tenenbaumet.alA.M.,DataStructuresUsingC&C++,Prentice Hall of India.</i> <i>*Latest editions of all the suggested books are recommended.</i>	
Course Code: BCACC240301	Discipline Core Course -3 BCA Semester -III OPERATING SYSTEM	L-3 T-0 P-0 C-3
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understanding to describe the functionality of operating system.	
CO-2	Understanding to understand process, its management and synchronization.	
CO-3	Understanding concept of deadlock, how it is detected and prevented.	
CO-4	Understanding concept of memory and its management and various processes to manage it.	
CO-5	Understanding concept of file management and its various techniques.	
Course Content		
Unit-1	Introduction to the Operating System (OS), Types of Operating System: Batch System, Time Sharing System, Real Time System. Multi Programming, Distributed System, Functions and Services of OS.	8 Hours
Unit-2	Process Management: Process Concept, Process State, Process Control Block, Process Scheduling, CPU Scheduling - Scheduling Criteria, Scheduling	8 Hours

	Algorithms, Preemptive & Non Preemptive Scheduling.	
Unit-3	Deadlocks: System model, Characterization, Deadlock Prevention, Deadlock Avoidance and Detection, Recovery from deadlock.	8 Hours
Unit-4	Memory Management: Logical Address, Physical Address, External and Internal Fragmentation. Concept of paging, Page table structure - Hierarchical Paging, Hashed Page Tables, Inverted Page Table.	8 Hours
Unit-5	Information Management: File Concept, Access Methods, Directory Structure. Device Management: Disk Structure, Disk Scheduling Algorithms.	8 Hours
Textbooks:	1. Silberschatz and Galvin, "Operating System Concept", Addison Wesley.	
Reference Books:	1. Flynn, Mchoes, "Understanding Operating System", Thomson Press, Third Edition. 2. Tannenbaum, "Operating System Concept", Addison Wesley. 3. Joshi, R. C. and Tapaswi, S., "Operating Systems", Wiley Dreamtech. 4. Nutt, G., "Operating Systems", Addison-Wesley. 5. GodboleAhyut, "Operating System", PHI. <i>Latest editions of all the suggested books are recommended</i>	
Course Code: BCACC240351	Compulsory Core Course LAB -3 BCA Semester-III Operating System Lab	L-0 T-0 P-2 C-1
CO1.	Understanding various methods to perform the basic operations on a UNIX based Operating system.	
CO2.	Evaluate the performance of different types of CPU scheduling algorithms.	
CO3.	Simulate Banker's algorithm for deadlock avoidance.	
CO4.	Use different system calls for writing application programs.	
CO5.	Ability to implement inters process communication between two processes.	
	LISTOFEXPERIMENTS	
	1. Write a shell program to add two numbers. 2. Write a shell program to find a number is even or odd. 3. Write a shell program for Fibonacci series. 4. Implementation of Process and thread (Life cycle of process): (i) Process creation and Termination; (ii) Thread creation and Termination 5. To implement CPU Scheduling Algorithms • FCFS • SJF • SRTF • PRIORITY • ROUNDROBIN 6. Simulate all Page Replacement Algorithms • FIFO • LRU 7. Simulate algorithm for deadlock prevention and detection 8. Simulate Paging Technique of Memory Management. 9. Write a C program for process creation and termination. 10. Write a C program for thread creation and termination. 11. Write a C program to simulate Bankers Algorithm for Deadlock Avoidance. Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.	
Course Code:	Discipline Core Course -4	L-3

BCACC240401	BCA Semester -IV DIGITAL ELECTRONICS AND COMPUTER ORGANIZATION	T-0 P-0 C-3
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understanding the basics of Number system, Boolean algebra and its applications in digital electronics.	
CO-2	Understanding different combinational and sequential circuits in digital electronics.	
CO-3	Understanding the organization of computer system and its components, memory hierarchy, I/O mechanism.	
CO-4	Applying the concepts to design various combinational and sequential circuits.	
CO-5	Analyzing the efficiency of various gates and flip-flops based upon their functionality.	
Course Content		
Unit-1	Number System: Data representation, Data Types and Number Systems, Binary Number System, Octal & Hexa-Decimal Number System; Fixed Point Representation; 1's&2's Complement; Binary Fixed Point Representation; Arithmetic Operation on Binary Numbers; Overflow & Underflow; Floating Point Representation; Codes: ASCII, EBCDIC Codes, Gray Code, Excess-3& BCD; Error Detection & Correcting Codes; Binary Storage and Registers.	8 Hours
Unit-2	Boolean algebra: Definition, Properties, Law's. Digital Logic Circuits: Logic Gates: AND, OR, NOT Gates and their Truth Tables, NOR, NAND & XOR Gates; Demorgan's Theorem; Map Simplification; Minimization Techniques: K Map Two, Three and More variables maps; Sum of Product & Product of Sums; Don't care conditions; Combination & Sequential Circuits; Half adder & Full adder; Full subtractor and decimal adder, Code Conversion; Multilevel NAND and NOR Circuits; Multiplexers and Demultiplexers; ROM Working & Circuit.	8 Hours
Unit-3	Sequential logic: Flip-flops: RS ,D, JK&T Flip-Flop ,Triggering in flip flops, Analysis of Clocked Sequential Circuits; State Reduction and Assignment; flip flop excitation tables; Design procedure and Design of Counters; Design with equations; Registers; Counters and the memory unit; Shift registers; Ripple counters and Synchronous counters; Timings sequence digital logic families; Processor organization; General Register Organization; Stack Organization and Addressing Modes.	8 Hours
Unit-4	Computer Registers and I/O: Register transfer logic; Intel Register Transfer; Arithmetic Logic and Shift Micro Operation; Conditional; Constant Statement; Fixed Point Binary Data Floating Point Data; Instruction Codes; Input-output organizations-I/O Interface; Properties of simple I/O Devices and their controller; Isolated vs. Memory-mapped I/O; Modes of data transfer; Synchronous & Asynchronous data transfer.	8 Hours
Unit-5	Computer Organization: Block Level Description of the Functional Units as Related to the Execution of a Program; Fetch; decode and Execute Cycle. Memory organization: Auxiliary Memory ;Magnetic Drum; Disk & Tape; Semiconductor memories; Memory; Hierarchy; Associative memory; Virtual memory; Address space & memory space; Address mapping; page table; Page replacement; segmentation; Cache memory; Hit ratio; Mapping techniques; Writing into Cache.	8 Hours
Textbooks:	1. Mano, M., Computer System Architecture, Prentice Hall of India.	
Reference Books:	1. Tannenbaum, O., Structured Computer Organization, Prentice Hall of India. 2. Hayes, P.J., Computer Organization, McGraw Hill. * Latest editions of all the suggested books are recommended.	
Course Code: BCACC240451	Compulsory Core Course LAB -4 BCA Semester IV DIGITAL ELECTRONICS AND COMPUTER ORGANIZATION Lab	L-0 T-0 P-2 C-1
Course	On completion of the course, the students will be:	

Outcomes:		
CO1.	Applying the concept of basic gates to design combinational circuits And verify using truth table.	
CO2.	Applying the concept of flip flops to design sequential circuits and Verify using truth table.	
CO3.	Applying the concepts to design registers.	
CO4.	Applying the concepts to design counters.	
CO5.	Applying the concepts of AND & OR Gate.	
	LIST OF EXPERIMENTS	
	➤ To study of following combination circuits: Multiplexer, Demultiplexer and Encoder. Verify truth tables of various logic functions. ➤ To study of various combinational circuits based on: AND/NAND Logic blocks and OR/NOR Logic blocks. ➤ To study various waveforms at different points of a transistor bi-stable multi-vibrator and its frequency variation with different parameters. ➤ To design a frequency divider using IC-555 timer. ➤ To study various types of registers and counters. ➤ To study Schmitt trigger circuit. ➤ To study transistor multi-vibrator. ➤ Experimental study of characteristics of CMOS integrated circuits. ➤ Interfacing of CMOS to TTL and TTL to CMOS. ➤ BCD to binary conversion on digital IC trainer. ➤ Testing of digital IC by automatic digital IC trainer. ➤ To study OPAMP as Current to Voltage & Voltage to Current converters & comparator.	
Course Code: BCACC240501	Discipline Core Course -5 BCA Semester -V SOFTWARE ENGINEERING AND TESTING	L- 3 T-0 P-0 C-3
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Applying the concepts of life cycle models to choose the appropriate model.	
CO-2	Analyzing the requirements and design the software.	
CO-3	Creating or implementing the software based on the industry standards	
CO-4	Designing and developing test cases.	
CO-5	Designing software by applying the software engineering principles.	
	Course Content	
Unit-1	Introduction: Software Engineering approach, Need of engineering aspect for Software Design, SDLC, Software Crisis, Software Process, Process models (Classical Waterfall Model, Build-n-Fix Model, Iterative Waterfall Model, Prototyping Model, Evolutionary Model and Spiral Model)	8 Hours
Unit-2	Software Requirement Analysis and Specifications: Software Requirement Specifications, Need of SRS, Steps for constructing good SRS, Behavioral and Non-Behavioral requirements, Analysis Model.	8 Hours
Unit-3	Software Design: Design Concepts & Principle, problem partitioning, abstraction, and top down and bottom up-design, Cohesion & Coupling, How to measure degree of Cohesion and Coupling, Function Oriented Design, DFDs, Structure Chart, Object Oriented Design. Coding: Top-Down and Bottom-Up programming, Structured programming, Programming style, Do's and Don'ts for Coding.	8 Hours
Unit-4	Coding: Programming languages and development tools, Selecting languages and tools, Good programming practices Coding Standards.	8 Hours
Unit-5	Software Maintenance: Software Maintenance Process and its types, Introduction	8 Hours

	to Reverse Engineering. Software Reliability & Quality Assurance: Software Reliability issues, Software quality, Overview of Quality Standards like ISO 9001, SEI-CMM and its comparison with ISO, Introduction, scope and architecture of CASE.	
Textbooks:	1. 1.RajibMall, "SoftwareEngineering", PHI..	
Reference Books:	1. P.Jalote,"AnIntegratedapproachtoSoftwareEngineering",Narosa. 2. WamanS.Jawadekar,"SoftwareEngineering:PrinciplesandPractice", 3. K.K.Agrawal&Yogesh Singh, "Software Engineering", New Age Publication. 4. R.S.Pressman,"SoftwareEngineering–Apractitioner'sapproach", 5. Ian Sommerville, Software Engineering , Pearson Education(Addison Wesley). * Latest editions of all the suggested books are recommended.	
CourseCode: BCACC240551	Compulsory Core Course LAB -5 BCA Semester V SOFTWARE ENGINEERING AND TESTING LAB	L-0 T-0 P-2 C-1
Course Outcomes:	On completion of the course, the students will be:	
CO1	Understand Software Development Life Cycle (SDLC)	
CO2	Apply Software Engineering Principles	
CO3	Implement Testing Techniques & using Testing Tools	
CO4.	Analyzing the methods for creating SRS and various diagrams using Software engineering paradigms.	
CO5.	Analyzing the methods for creating flow chart to show process flow.	
	LISTOFEXPERIMENTS	
	For any given case/problem statement do the following; 1. Prepare a SRS document in line with the IEEE recommended standards. 2. To perform the function oriented diagram: Data Flow Diagram (DFD) and Structured chart. 3. Draw the use case diagram and specify the role of each of the actors. 4. Also state the pre-condition ,post condition and function of each use case. 5. Draw the activity diagram. 6. To draw the structural view diagram for the system: Class diagram, object diagram. 7. Identify the classes Classify them as weak and strong classes and draw the class diagram. 8. Draw this quince diagram for any two scenarios. 9. Draw the collaboration diagram. 10. Draw the state chart diagram. 11. Draw the component diagram. 12. To perform various testing using the testing tool unit testing, integration testing for a sample code of the suggested system. 13. Perform forward engineering in java.(Model to code conversion) 14. Perform reverse engineering in java.(Code to Model conversion) Note: The Instructor ma add/delete/modify/tune experiment, wherever he/she feels in a justified manner.	
Course Code: BCACC240601	Discipline Core Course -6 BCA Semester -VI COMPUTER NETWORK	L-3 T-0 P-0 C-3

Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understanding the basics of computer network, topology, data communication system and availability of information.	
CO-2	Understanding the layers of the OSI model and TCP/IP and explain the function of each layer.	
CO-3	Analyzing the different types of networking protocol and their functionality.	
CO-4	Analyzing the services and features of the various layers such as Application, Transport of data networks.	
CO-5	Applying the various protocols studied at different layers in real life application.	
Course Content		
Unit-1	Introductory Concepts: Goals and Applications of Networks, Network structure and architecture, the OSI reference model, services, networks topology, Physical Layer- transmission, switching methods, Integrated services digital networks.	8 Hours
Unit-2	Medium access sub layer: Channel allocations, LAN protocols, ALOHA Protocols- Pure ALOHA, slotted ALOHA, Carrier Sense Multiple Access Protocols, CSMA with Collision Detection, Collision free Protocols, IEEE standards, Ethernet, Error correction & detection algorithms, elementary data link layer protocols, sliding window protocols, error handling, High Level Data Link Control.	8 Hours
Unit-3	Network Layer: Point-to Point networks, concept of virtual circuit and LAN, routing algorithms, congestion control algorithms, internetworking, TCP/IP protocol, UDP, SCTP, IP addresses, IPv6 Packet Format, Subnetting.	8 Hours
Unit-4	Transport Layer: Design issues, connection management, Internet Transport Protocol (UDP), Ethernet transport Protocol, Transmission Control Protocol (TCP).	8 Hours
Unit-5	Application Layer: Domain Name System, Simple Network Management Protocol, Electronic mail, File Transfer Protocol, Hyper Text Transfer Protocol, Introduction to Cryptography and Network Security Communication Security (IPSec, Firewalls). Implement a program for encryption a text with network security algorithm.	8 Hours
Textbooks:	1. Computer Networks by A. S Tanenbaum”, Pearson education.	
Reference Books:	1. Computer Networks with Internet Protocols by W Stallings, Pearson Education. 2. Local and Metropolitan Area Networks by W Stallings, Pearson 3. Data and Computer Communication by W. Stallings, Macmillan Press. 4. Computer Networks & Internet with Internet Applications by Comer Pearson Education. 5. Internetworking with TCP/IP by PHI. 6. Data Communication and Networking by Forouzan TMH. * Latest editions of all the suggested books are recommended.	
Course Code: BCACC240651	Compulsory Core Course LAB -6 BCA Semester IV COMPUTER NETWORK LAB	L-0 T-0 P-2 C-1
Course Outcomes:	On completion of the course, the students will be:	
CO1.	Understanding the working of network simulation tool (Packet Tracer)	
CO2.	Understanding about basic network connectivity .Understand IOS used for Networking devices	
CO3.	Understanding about ARP table. Analyzing some trouble shooting commands	
CO4.	Applying the knowledge to Configure the initial switch and router setting, Understand TCP/IP and OSI models	
CO5.	Analyzing MAC and IP addresses, Learn about TCP and UDP communications.	
LIST OF EXPERIMENTS		
1. To write a program in C Illustrating use of TCP Sockets. 2. To write a program in C illustrating use of simple UDP.		

	3. To write a program in C illustrating use of Raw Sockets (like packet capturing and filtering). 4. To write a program in C illustrating concept of Sliding Window Protocol. 5. To write a program in C for Address Resolution Protocol. 6. To write a program in C for implementing Routing Protocols. 7. To write a program in C illustrating for Open Shortest Path first Routing Protocol.	
Group A: Artificial Intelligence (AI)		
Course Code: BCAAI240101	Group A : Artificial Intelligence (AI) Discipline Major/Minor Course -1 BCA Semester -I ARTIFICIAL INTELLIGENCE	L-4 T-0 P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	To understand the concept of classify different knowledge and their representation	
CO-2	To understand the concept about intelligence and different perspectives of researchers about artificial intelligence.	
CO-3	To understand the concept about expert systems, the process of their creation and usage.	
CO-4	To understand the concept of intelligent agents, their structure and behavior.	
CO-5	To understand the concept about various domains where artificial intelligence could be used.	
Course Content		
Unit-1	Introduction–Definition of Hypothesis, knowledge, knowledge type’s, knowledge representation using propositional logic , predicate logic.	8 Hours
Unit-2	Introduction to Intelligence; Intelligent behavior ;Various definitions ofArtificialIntelligence;HistoryofAI;Goals ofArtificialIntelligence,PartsofArtificia lIntelligence.	8 Hours
Unit-3	Introduction, Knowledge Base, Working Memory, Inference Engine, Expert System ,Expert System Shells ,Application of Expert Systems i.e. Mycin	8 Hours
Unit-4	Definition of Intelligent Agents; Software Agents Architecture for Intelligent Agents; Characteristics of Intelligent Agents; Types of Intelligent Agents; Application of Intelligent Agents.	8 Hours
Unit-5	ApplicationsAIapplications– InformationRetrieval;InformationExtraction;NaturalLanguageProcessing;Machi neTranslation;SpeechRecognition;Robot–Hardware–Perception–Planning–Moving	8 Hours
Textbooks:	1. S.RussellandP.Norvig,"ArtificialIntelligence:AModernApproach,PrenticeHall,ThirdEdit ion, 2009. 2. ArtificialIntelligence:AModernApproach,4thEdition,StuartRussell,peterNorvigUniversit yofCalifornia at Berkeley,Pearsoneducation, 2020. I.Bratko,—Prolog:ProgrammingforArtificialIntelligence,FourthEdition,Addison- WesleyEducationalPublishersInc., 2011	
Reference Books:	1. Cloksin,W.F.,Mellish,C.S., <i>ProgrammingInProlog</i> ,NarosaPublishingHouse. 2. Janakiraman,V.S.,Sarukesi,K., <i>Foundationof ArtificialIntelligence&ExpertSystem</i> ,Macmillan. * Latest editions of all the suggested books are recommended.	
Course Code: BCAAI240201	Group A : Artificial Intelligence (AI) Discipline Major/Minor Course -2 BCA Semester -II Introduction to Machine Learning	L-4 T-0 P-0 C-4

Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understanding basic concept of machine learning, advantages and disadvantages, applications, learning algorithms: supervised learning, unsupervised learning, semi- supervised learning, reinforcement learning,decision trees, Hunt’s algorithm for learning a decision tree.	
CO-2	Understanding concept of KNNs, SVMs and Naïve Bayes algorithms in text classification, decision boundary of KNN, feature selection using KNN, linear classifiers.	
CO-3	Understanding concept of ANN and regression, perceptron algorithm, decision boundary of single neuron, linear regression, logistic regression, and logistic regression for multi-class classification.	
CO-4	Applying concept of feature selection and feature extraction, filter based methods for feature selection, wrapper methods for features selection.	
CO-5	Applying concept of sequence labeling and clustering in classification, probabilistic sequence model, hidden markov model in classification, K-mean clustering, hierarchical clustering methods.	
Course Content		
Unit-1	Introduction to Machine Learning: Introduction; History, Advantages, Scope and Applications; Learning Algorithms: Supervised Learning; Unsupervised Learning; Semi-Supervised Learning; Reinforcement Learning. Decision Trees: Introduction, Scope, Advantages; Hunt's algorithm for learning a decision tree; Details of tree induction;	8 Hours
Unit-2	KNNs, SVMs and Naïve Bayes: Examples of few text classification problems; Naïve Bayes for text classification; Introduction to KNN algorithm; Decision boundary KNN Vs Decision tree; What is the best K; KNN Problems; Feature selection using KNNs; Linear Classifiers; Learning non-linear patterns.	8 Hours
Unit-3	ANN and Regression: Motivation for Artificial Neural Network; Perceptron Algorithm; Decision Boundary for a single Neuron; Introduction to Linear Regression; Coefficient of Determination; Logistic regression vs Linear Regression; Can we use Regression Mechanism for Classification?; Logistic Regression – Deriving the Formula; Logistic Regression for Multi-class Classification; Logistic Regression Decision Boundary.	8 Hours
Unit-4	Feature Selection: Introduction tofeature selection: what, why, how and where?; Feature selection vs feature extraction; Feature subset selection using Filter based methods; Wrapper Methods; Wrapper Methods vs Filter Methods; Model based machine learning with regularization; Regularization using L2 and L1.	8 Hours
Unit-5	Sequence Labeling, Clustering: Introduction to Sequence Learning; Sequence Labeling as Classification; Probabilistic Sequence Models; Hidden Markov Model; Three Problems of an HMM. Basics of Clustering; Applications of Clustering; Understanding Distance based Clustering; K-means Algorithm; Hierarchical (Agglomerative) clustering; Evaluation of cluster quality.	8 Hours
Textbooks:	1. Brian, O, Management Information System, Tata McGraw Hill.	
Reference Books:	1. Brian, O., Introduction to Information System, McGraw Hill. 2. Jawadekar, W., Management Information System, Tata McGraw Hill. 3. Jain, S., Management Information System, Tata McGraw Hill. * Latest editions of all the suggested books are recommended.	
Group B: Data Analytics (DA)		
Course Code: BCADA240101	Group B: Data Analytics (DA) Discipline Major/Minor Course -4	L- 4 T-0

	BCA Semester -I FUNDAMENTAL OF DATA ANALYTICS TECHNIQUES	P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understanding the various types of data	
CO-2	Comparing the data analytics with other related terms	
CO-3	Understanding the various data pre-processing methods	
CO-4	Learning various techniques of descriptive analytics	
CO-5	Learning various techniques of predictive analytics	
Course Content		
Unit-1	Data to Data Science, Understanding data, Type of Data: Numeric, Categorical, Graphical, High Dimensional Data, Data Classification, Data Formats, Structured, Semi-Structured and Un-Structured, Data Sources, Time Series, Transactional Data, Biological Data, Spatial Data ,Social Network. Data Evolution, ICT Advancement & data growth -a Perspective,	8 Hours
Unit-2	Data Analytics vs Statistics, Data Analytics vs Data Science, Data Science vs Database, Data Analytics vs Machine Learning. What is Big Data – Sources of Big Data, Data Analytics Relation: Data Science, Analytics and Big Data Analytics. Data Science Components, Big data technology, Data Science user- roles and skills.	8 Hours
Unit-3	Data Collection Strategies, Types of data, Data Pre-Processing, Data Cleaning, Data Integration and Transformation, Data Reduction, Data Discretization.	8 Hours
Unit-4	Descriptive Analysis: Measure of central tendency: Mean, Median, Mode, Skewness and Kurtosis, Box Plots, Pivot Table, Heat Map, Correlation Statistics, Simple and Multiple Regression, Model Evaluation using Visualization,	8 Hours
Unit-5	Predictive analysis : Regression analysis, Classification techniques ,Outlier Analysis, Clustering, Association rules analysis	8 Hours
Textbooks:	<i>1 Joel Grus, “Data Science from Scratch”, 2nd Edition, O'Reilly Publisher</i>	
Reference Books:	<i>1 Cathy O’Neil and Rachel Schutt , “Doing Data Science”, O'Reilly, 2015.</i> <i>2 SinanOzdemir, Sunil Kakade, “Principles of Data Science”, Second Edition 3Field Cady, “The Data Science Handbook”, John Wiley & Sons, Publishers,</i> * Latest editions of all the suggested books are recommended.	
Course Code: BCADA240201	Group B: Data Analytics (DA) Discipline Major/Minor Course -5 BCA Semester -II Probability & Statistics	L- 4 T-0 P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understand the fundamentals of probability, Conditional Probability, Baye’s theorem,	
CO-2	Understand the fundamentals of probability distribution and other distributions	
CO-3	Understanding fundamentals of statistics	
CO-4	Understanding the concept of correlation and regression	
CO-5	Perform hypothesis testing to analyse various problems.	
Course Content		
Unit-1	Basic terminology-random experiment, sample space, event, mutually exclusive events, equallylikely events.Definition of probability, Classical, empirical and axiomatic approaches. Addition rule for two events, Independence of events, conditional probability, Multiplication rulefor two events.Bayes’ theorem and its applications.	8 Hours
Unit-2	Definition of a random variable—discrete and continuous random variables—probability massfunction and probability density function. Distribution function and its properties.Mean and variance of a random variable.Definition of a bivariate probability distribution and marginal probability functions.Discrete probability distributions—Binomial and Poisson distributions, Continuous distributions—	8 Hours

	Normal distribution—computation of probabilities	
Unit-3	Basic statistical concepts, qualitative and quantitative data, Classification of data, Construction of frequency distribution, stem and leaf display. Graphical presentation of data -histograms and cumulative frequency curves.Measure of central tendency—Arithmetic mean, median and mode, Standard deviation, Partition values—quartiles, deciles and percentiles. Measures of dispersion—Range, quartile deviation and standard deviation and their relativemeasures. Skewness—concept and measures, Kurtosis	8 Hours
Unit-4	Correlation—definition, scatter plot, types of correlation, measures—Karl Pearson’s correlation coefficient and Spearman’s rank correlation coefficient. Regression---Linear regression and interpretation	8 Hours
Unit-5	Statistical hypothesis—null, alternative, simple and composite.Types of errors, level of significance and power of a test.Tests of significance—Test for normal mean and equality of means for large samples. Chi square test for independence of attributes and goodness of fit.	8 Hours
Textbooks:	Walpole, Myers et al. –Probability and statistics for scientists and engineers., PearsonEducation	
Reference Books:	J K Sharma, Business Statistics, Vikas Publishing Agarwal B.L, Basic Statistics New age international, Runger and Montgomery, Applied Statistics and probability for Engineers, Wiley * Latest editions of all the suggested books are recommended.	
Group C: Cyber Security (CS)		
Course Code: BCACS240101	Group C: Cyber Security (CS) Discipline Major/Minor Course -6 BCA Semester -I Fundamentals of Cyber Security	L- 4 T-0 P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understand the basic concepts and importance of cybersecurity and cyberspace.	
CO-2	Identify and classify various types of cyber crimes and network attacks.	
CO-3	Analyze social media platforms, their security concerns, and best practices for safe usage.	
CO-4	Understand the legal framework governing cyber crimes and reporting mechanisms.	
CO-5	Comprehend the components and security aspects of e-commerce and digital payments.	
Course Content		
Unit-1	Introduction to Cyber security; Defining Cyberspace and Overview of Computer and Web-technology, Communication and web technology, Internet, World wide web, Advent of internet, Regulation of cyberspace, Concept of cyber security, Issues and challenges of cyber security.	8 Hours
Unit-2	Introduction to Cyber crime and Network attacks; Classification of cyber crimes, Common cyber crimes- cyber crime targeting computers and mobiles, cyber crime against women and children, financial frauds, social engineering attacks, malware and ransomware attacks. Firewalls and Packet Filters, password Cracking, Keyloggers and Spyware, Virus and Worms, DOS and DDOS attack, Buffer Overflow, Attack on wireless Networks.	8 Hours
Unit-3	Social Media Overview and Security: Introduction to Social networks. Types of Social media, Social media platforms, Social media monitoring, Hashtag, Viral content, Social media privacy, Challenges, opportunities and pitfalls in online social network. Flagging and reporting of inappropriate content on Social media, Best practices for the use of Social media, Case studies.	8 Hours
Unit-4	Reporting of cyber crimes and Cyber Law: Remedial and mitigation measures, Legal perspective of cyber crime, IT Act 2000 and its amendments, Cyber crime and offences, Organisations dealing with Cyber crime and Cyber security in India, Case studies.	8 Hours

Unit-5	E - Commerce and Digital Payments: Components of E-Commerce, Elements of E-Commerce security, E-Commerce threats, E-Commerce security best practices, Introduction to digital payments, Components of digital payment and stake holders, Modes of digital payments- Banking Cards, Unified Payment Interface (UPI), e-Wallets, Aadhar enabled payments, Digital payments related common frauds and preventive measures.	8 Hours
Textbooks:	1. Fundamentals of Cyber Security, Mayank Bhushan, Rajkumar Singh Rathore, Aatif Jamshed 2. Cybersecurity Essentials, Charles J. Brooks, Christopher Grow, Philip Craig, Donald Short	
Reference Books:	1. Cybersecurity and Cyberwar: What Everyone Needs to Know, P.W. Singer, Allan Friedman 2. The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws, Dafydd Stuttard, Marcus Pinto <i>*Latest editions of all the suggested books are recommended.</i>	
Course Code: BCACS240301	Group C: Cyber Security (CS) Discipline Major/Minor Course -7 BCA Semester -II Cyber Crime & Cyber Law	L- 4 T-0 P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understand the fundamental concepts of cybercrime and its impact on information security.	
CO-2	Identify and explain the different types of cyber crimes and the legal frameworks associated with them.	
CO-3	Understand about cyber laws, legislative and judicial frameworks, and basic legal terms and concepts.	
CO-4	Analyze law and policy in Indian cyberspace, including jurisdiction and regulatory issues.	
CO-5	Understand cyber torts, intellectual property issues, and their interface with copyright and patent laws.	
Course Content		
Unit-1	Introduction to Cybercrime: Cybercrime definition and origins of the world, Cybercrime and information security, Classifications of cybercrime, Common cyber-crimes- cyber-crime targeting computers and mobiles, cyber-crime against women and children, financial frauds, social engineering attacks ,Cybercrime and the Indian ITA 2000, A global Perspective on cybercrimes.	8 Hours
Unit-2	Cyber Crimes & Legal Framework: Cyber Crimes against Individuals, Institution and State, Hacking, Digital Forgery, Cyber Stalking/Harassment, Cyber Pornography, Identity Theft & Fraud, Cyber terrorism, Cyber Defamation, Different offences under IT Act, 2000.	8 Hours
Unit-3	Introduction to Cyber Laws: Introduction to Cyber Law, Legislative & Judicial Framework, Basic legal terms and concepts, Computer Software & Copyright Law , Trademark Issues in Cyberspace, Software Licenses, Computer Databases & Law, Cyber Frauds, Cyber Pornography,Cyber Terrorism, Data Privacy & Confidentiality, Electronic & Digital Signatures – Legal Issues.	8 Hours
Unit-4	Law and Policy in 'Indian' Cyberspace: Jurisdiction in Cyberspace, Designing On-Line Agreements: Contractual Closure and Indian Contract Law Electronic Signatures & Digital Signatures, Regulation of Certifying Authorities, Electronic Evidence, Data Protection, Privacy and Corporate Compliance, Intellectual Property, The Internet and Electronic Commerce, Managing Legal Risks when promoting a business online, The Legal Challenges to Internet Banking.	8 Hours
Unit-5	Cyber Torts: Different Types of Civil Wrong sunder the IT Act, 2000, Intellectual Property Issues in Cyber Space, Interface with Copyright Law, Interface with Patent Law, Trade marks & Domain Names Related issues.	8 Hours

Textbooks:	1. <i>Cyber Law: The Law of the Internet and Information Technology</i> , Anirudh Rastogi 2. <i>Cybercrime and Cybersecurity in the Global South</i> , Charlette Donalds, Gloria M. Vibert	
Reference Books:	1. <i>Cyber Law in India</i> , Farooq Ahmad 2. <i>Criminal Threats from Cyberspace</i> , Susan W. Brenner <i>*Latest editions of all the suggested books are recommended.</i>	
Group D: Full Stack (FS)		
Course Code: BCAFS240101	Group D : Full Stack (FS) Discipline Major/Minor Course -8 BCA Semester -I JAVA	L- 4 T-0 P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understanding the detail concept of java in real life.	
CO-2	Learning how the java is different and easy from other programminglanguages.	
CO-3	Analyze the relationship between java and data analysis.	
CO-4	Understanding java with some modules.	
CO-5	Understanding how the data is predicted in java.	
Course Content		
Unit-1	An Introduction to Java: Introduction to Java Concept of class, Introduction of Object Oriented Programming concept, Features of JAVA, Basic concepts of OOPs: Object, Class, Inheritance, Polymorphism, Abstraction, Encapsulation, Structure of JAVA Program ,Creating a JAVA source file ,Compiling of JAVA source file and Compilation process ,Run the application using JAVA interpreter. Using Data with JAVA Programs: Constants, Literals, Variables, Keywords, Identifier, Data types (Integers, Floating Point, Character, Boolean) .Operators(Arithmetic, Relational, Boolean, Increment and decrement, conditional, Bitwise)	8 Hours
Unit-2	Conditional statements, Array and string : Control Statements, Flow Control Statements(If and if else , Nesting IF ... else , Switch case statement) ,Looping Statement(While loop, for loop, do..while loop, Nested Loop) Array and String in JAVA : Declaring and initializing an array ,Using two dimensional and multidimensional arrays , Passing arrays to methods , methods of arrays fill(), sort(), equal(),binary search() , Initialization of string, Manipulating string class (isUppercase(), toUppercase(), isLowercase(), toLowercase()) String methods	8 Hours
Unit-3	Methods , Classes and Objects : Class variables and class methods ,Classification of variable declared in a class(local variable, Instance variable,,Class variable) , Method Overloading , Constructors , Visibility modifiers for Access control(Public, Private , Protected) ,Passing object as a parameter in a method , Inheritance ,Super class ,Final method, Static method, Abstract class.	8 Hours
Unit-4	Multithreaded Programming and Package, Interface : Use of Multithread programming, Thread class and Runnable interface ,Thread priority, Thread synchronization, Thread communication, Deadlock, Use of Package, CLASSPATH, Import statement, Static import, Access control, Creation and Implementation of an interface, Interface reference, instance of operator, Interface inheritance, Dynamic method dispatch ,Comparison between Abstract Class and interface	8 Hours
Unit-5	Exception Handling: Exception Type, Using try and catch, multiplecatch Clauses, Nested try Statements, Using throw keyword, Using throws keyword, Using Finally, Chained Exception. Input / Output: I/O Basic, The Stream Classes, Reading Console Input, WritingConsole Output.	8 Hours

Textbooks:	1. Patrick Naughton & Herbert Schildt, The Complete Reference JAVA2, Tata McGraw Hill	
Reference Books:	1. Balagurusamy E., Programming in JAVA, Tata McGraw Hill. 2. Steven Holzner, Java2 Black Book, Dreamtech. 3. Mark Wutica, —Java Enterprise Edition I, QUE. 4. Cay S. Horstmann & Gary Cornell, Core Java2 Volume I—Fundamentals, PHI *Latest editions of all the suggested books are recommended.	
Course Code: BCAFS240201	Group D : Full Stack (FS) Discipline Major/Minor Course -9 BCA Semester -II Agile Technology	L- 4 T-0 P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Apply Agile requirement techniques to streamline software development processes	
CO-2	Analyse various Agile software methodologies to identify the most suitable approach for the project	
CO-3	Evaluate different Agile estimation techniques to accurately plan project timelines and resource allocation	
CO-4	Design an Agile testing approach to ensure the quality and functionality of software products throughout the development lifecycle.	
Course Content		
Unit-1	Introduction to Agile Methodologies : Traditional approach of Software Development Methodology, Need of Agile software Development, Defining Agile, Agile Manifesto Principles of Agile , Values of Agile ,Business Benefits of Agile Software Development , Traditional Requirements Development , Agile Requirements : Epics and User stories ,Difference between Epics and User stories ,Backlog Management, Class Responsibility Collaborator.	8 Hours
Unit-2	Scrum and Kanban Methodologies: Introduction to Scrum framework, Advantages of Scrum Framework. Phases of Scrum, Principles of Scrum, Roles: Product owner, team members and scrum master, Scrum Ceremonies :Sprint, sprint planning, daily scrum, sprint review, and sprint retrospective, Artifacts: Product backlog, sprint backlog and increments. Introduction to Kanban framework , Workflow, Limit the amount of work in progress, pulling work from column to column, Kanban board, Adding policies to the board, Cards and their optimization.Kanban Practices , Kanban Flow practices.Work Item Age. Kanban vs Scrum.	8 Hours
Unit-3	Extreme Programming and Agile Estimation Techniques: Basic values and principles, Roles, Twelve practices of XP, Pair programming, XP team, Life cycle and tools for XP., Good practices need to be practiced in extreme programming, Advantages of Extreme Programming, Agile Maturity Model and Agile Estimation Techniques - Planning Poker-Shirt Sizes. Dot Voting, Bucket System.	8 Hours
Unit-4	Agile Testing: Agile Testing Life Cycle, Agile Testing Quadrants, Agile Testing Techniques: Behavior Driven Development, Test Driven Development Acceptance Test Driven Development Testing.Role of Agile Tester.User stories approach in Acceptance Test Driven Development Testing.Other Techniques - Exploratory Testing , Session Based testing, Agile Test Metrics.	8 Hours
Unit-5	Agile Project Management: Agile vs. Traditional Project Management,Key Differences and Benefits of Agile, Role of the Scrum Master, Facilitating Teams and Removing Obstacles.Agile Metrics and Reporting, Tracking Progress and	

	Performance.	
Textbooks:	1. Stellman, A., & Hart, J. A. (2015). Learning Agile. O'Reilly	
Reference Books:	1. Crispin, L., & Gregory, J. (2008). Agile Testing: A Practical Guide for Testers and Agile Teams. Addison Wesley. 2. Schwaber, K., & Beedle, M. (2002). Agile Software Development with Scrum. Pearson. 3. Martin, R. C. (2002). Agile Software Development, Principles, Patterns and Practices. Pearson. *Latest editions of all the suggested books are recommended.	
	Group E: Web Applications (WA)	
Course Code: BCAWA240101	Group E: Web Applications (WA) Discipline Major/Minor Course -10 BCA Semester -I Django	L- 4 T-0 P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understand the fundamentals of web development and the Django framework	
CO-2	Develop Django models and integrate them with views and templates	
CO-3	Implement user authentication and authorization in Django	
CO-4	Develop Django forms for data input and validation	
CO-5	Design and deploy scalable Django applications	
	Course Content	
Unit-1	Introduction to Django: Overview of Web Development: Introduction to web frameworks, Django's role in web development. Setting Up Django: Installation, virtual environments, basic project structure. Creating a Simple Application: Setting up a basic Django project, understanding the MVC (Model-View-Controller) architecture in Django (also known as MTV - Model-Template-View)	8 Hours
Unit-2	Django Models and Admin Interface: Django Models: Defining models, relationships between models, Django ORM (Object-Relational Mapping). Admin Interface: Using Django's built-in admin interface for CRUD (Create, Read, Update, Delete) operations.	8 Hours
Unit-3	Django Views and Templates: Views and URLs: Creating views, URL routing, class-based views vs function-based views. Templates: Using Django template language (DTL), template inheritance, including templates, template filters and tags.	8 Hours
Unit-4	Django Forms and Authentication: Django Forms: Creating forms in Django, handling form submission, form validation. User Authentication: User authentication system in Django, login/logout functionality, user permissions and groups.	8 Hours
Unit-5	Advanced Django Concepts and Project Development: Middleware: Understanding middleware in Django, creating custom middleware. Handling Files: File uploads and handling in Django. Testing and Debugging: Testing Django applications, debugging techniques. Project Development: Collaborative project development using Django, integrating front-end frameworks (like Bootstrap) with Django.	8 Hours

Textbooks:	1. "Django for Beginners" by William S. Vincent 2. "Django 3 By Example" by Antonio Mele	
Reference Books:	1. "Django for APIs: Build web APIs with Python & Django" by William S. Vincent. 2. "Django Design Patterns and Best Practices" by ArunRavindran 3. "Test-Driven Development with Python: Obey the Testing Goat: Using Django, Selenium, and JavaScript" by Harry J.W. Percival <i>* Latest editions of all the suggested books are recommended.</i>	
Course Code: BCAWA240201	Group E: Web Applications (WA) Discipline Major/Minor Course -11 BCA Semester -II PHP with MySQL	L- 4 T-0 P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understand the concept of web servers, server side and client side scripts on web development.	
CO-2	Understand the basics of PHP language, its syntax, Wordpress and CodeIgniter framework, AJAX.	
CO-3	Understand the basics of MySQL, creating database and database connectivity in PHP with MySQL.	
CO-4	Apply concepts of PHP language using XML and AJAX to create more interactive web applications.	
CO-5	Apply MySQL with PHP to create dynamic content that interact with database.	
Course Content		
Unit-1	Web Servers: Introduction to web server. Installation and Configuration of XAMPP, CodeIgniter and CackPHP. An Introduction to server side scripts. Role of client side script and server side scripts on web development. Difference of PHP with other server side languages.	8 Hours
Unit-2	PHP- Introduction to PHP, History, Web Server, WAMP, Installation and Configuration files, Syntax, Operators, Variables, Constants, Control, tructure, Language construct and functions, Function – Syntax, Arguments, Variables, References, Returns and Variable Scope.	8 Hours
Unit-3	Arrays and its types, Date and Time functions, OOP ‘s – Installation, Modifiers, Inheritance, Interfaces, Exceptions, Static Methods and Properties, String functions. Web Features- Sessions, Forms, GET and POST data, Cookies.	8 Hours
Unit-4	Introduction to MySQL, Putting password on root. Creating users on MySQL. Creating database in MySQL. Creating table and using Sql queries in MySQL .Database connectivity in PHP with mysql_connect and PDO and using, insertion, deletion, updating and retrieval of data from database using PHP.	8 Hours
Unit-5	Introduction to AJAX, AJAX and its applications, working of AJAX, Introduction to web services, advantages of web services, web services platform elements, Using XML in PHP. Understanding the WordPress and CakePHP framework of PHP.	8 Hours
Textbooks:	1. Ivan Bayross, HTML, DHTML, JavaScript, CSS, PHP, BPB Publications.	
Reference Books:	1. PHP: The Complete Reference McGraw Hill Professional, 2020 <i>* Latest editions of all the suggested books are recommended.</i>	
Group F: Software Engineering & Quality Testing (ST)		
Course Code: BCAST240101	Group F: Software Engineering & Quality Testing (ST) Discipline Major/Minor Course -12 BCA Semester -I INTRODUCTION TO SELENIUM	L- 4 T-0 P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understanding the basics of Selenium and its components.	

CO-2	Understand and Discuss the Basic concepts of OOPs and Introduction of Core Java	
CO-3	Apply Selenium Commands and Develop Test Cases & Test Suites with Selenium-IDE	
CO-4	Understand working of Web Driver and Identify Web Element using different parameters, Apply Verification and Validation Commands.	
CO-5	Understanding Testing framework and Creating Test Suit with TestNG.	
Course Content		
Unit-1	Selenium Introduction - Selenium Introduction , What is Selenium and Why Selenium , Use of Automation Testing? ,Explanation of Selenium and its advantages , Differences between Selenium and QTP ,Introduction of selenium Components. .	8 Hours
Unit-2	OOPs concepts and Core Java Introduction - OOPs concepts , Setup Java environment – Download JDK and Setup eclipse , Java programming concepts, Class, Object, variables, methods, Packages ,Conditions and loops ,Access and non-access modifiers ,Advanced Java Programming ,Interfaces, Inheritance ,Collections, Exceptions.	8 Hours
Unit-3	Selenium IDE and RC Introduction - Installing Selenium IDE ,“Selenese” – Selenium Commands ,Actions, Asserts, Assessors. ,Developing Test Cases & Test Suites with Selenium-IDE ,Introduction to Selenium RC.	8 Hours
Unit-4	Selenium WebDriver 2.0 and 3.0 Introduction - Selenium WebDriver Introduction ,WebDriver Vs RC ,Download and Configure WebDriver with Eclipse ,Simple Testcase ,Open and Close Browser ,Cross Browser Testing – Firefox, IE, Chrome, Edge. ,UI elements Locators. ,IdentifyingWebElement using id, name, link text,PartialLinkText class Name, ,XPath, CssSelector and TagName,Verification Commands,Validation commands	8 Hours
Unit-5	Testing Framework and Tools - Introduction of the Testing framework ,Types of frameworks ,Tools for developing Test Framework ,TestNG introduction and Configuration with eclipse ,TestNG Annotations and Data Providers ,Creating Test Suit with TestNG.	8 Hours
Textbooks:	1. <i>Selenium WebDriver 3 Practical Guide" by UnmeshGundecha</i> 2. <i>"Mastering Selenium WebDriver 3.0" by Mark Collin</i> 3. <i>"Selenium WebDriver Recipes in Java" by Zhimin Zhan</i>	
Reference Books:	1. <i>"Selenium Testing Tools Cookbook" by UnmeshGundecha</i> 2. <i>"Learning Selenium Testing Tools with Python" by UnmeshGundecha and Rahul Shetty</i> * <i>Latest editions of all the suggested books are recommended.</i>	
Course Code: BCAST240201	Group F: Software Engineering & Quality Testing (ST) Discipline Major/Minor Course -13 BCA Semester -II SOFTWARE TESTING: PRINCIPLES, TECHNIQUES AND TOOLS	L- 4 T-0 P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understand the SDLC Process of a software and Describe terminology used in developing software.	
CO-2	Discuss SRS and its methods used for Verification and Source code reviews. understand various Software Quality Assurance Program.	
CO-3	Understand different testing techniques and Apply them with the help of examples.	
CO-4	Understand Regression Testing and Software Testing Activities Apply them with the help of examples.	
CO-5	Illustrate the principles and concepts of Automated Test Data Generation for solving problems related to software engineering.	
Course Content		

Unit-1	Review of Software Engineering: Overview of software evolution, SDLC, Testing Process, Terminologies in Testing: Error, Fault, Failure, Verification, Validation, Difference between Verification and Validation, Test Cases, Testing Suite, Test Oracles, Impracticality of Testing All data; Impracticality of testing All Paths.	8 Hours
Unit-2	Verification: Verification methods, SRS verification, Source code reviews, User documentation verification, Software project audit, Tailoring Software Quality Assurance Program by Reviews, Walkthrough, Inspection, and Configuration Audits.	8 Hours
Unit-3	Functional Testing: Boundary Value Analysis, Equivalence Class Testing, Decision Table Based Testing, Cause Effect Graphing Technique. Structural Testing: Control flow testing, Path testing, Independent paths, Generation of graph from program, Identification of independent paths, Cyclomatic Complexity, Data Flow Testing, Mutation Testing.	8 Hours
Unit-4	Regression Testing: What is Regression Testing? Regression Test cases selection, Reducing the number of test cases, Code coverage prioritization technique. Software Testing Activities: Levels of Testing, Debugging, Testing Techniques and their applicability, Exploratory Testing.	8 Hours
Unit-5	Automated Test Data Generation: Test Data, Approaches to test data generation, test data generation using genetic algorithm, Test Data Generation Tools, Software Testing Tools, and Software test Plan	8 Hours
Textbooks:	1. Yogesh Singh(2012), “Software Testing”, Cambridge University Press, New York, 3rd edition..	
Reference Books:	1. Roger S. Pressman(2001), “Software Engineering – A Practitioner’s Approach”, Fifth edition., McGraw-Hill International edition., New Delhi. 2. Marc Roper(2012), “Software Testing”, McGraw-Hill Book Co., London, 5th edition.. 3. Alexis Leon, Methews Leon, (1999), “Fundamentals of Information Technology”, Vikas Publishing, New Delhi. * Latest editions of all the suggested books are recommended.	
Group G: Cloud Computing and IOT (CI)		
Course Code: BCACI240101	Group G: Cloud Computing and IOT (CI) Discipline Major/Minor Course -14 BCA Semester -I Introduction To Cloud Computing	L- 4 T-0 P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understand the core concepts and architecture of cloud computing and its benefits.	
CO-2	Explain and differentiate between various cloud service models (IaaS, PaaS, SaaS) and deployment models.	
CO-3	Implement virtualization techniques and understand their role in cloud computing.	
CO-4	Apply resource management techniques to optimize cloud service usage.	
CO-5	Identify and address security, privacy, and legal issues in cloud environments.	
Course Content		
Unit-1	Introduction to Cloud Computing Definition of Cloud Computing, History and Evolution of Cloud Computing, Essential Characteristics of Cloud Computing, Cloud Computing Architecture: Frontend and Backend, Benefits and Challenges of Cloud Computing	8 Hours
Unit-2	Cloud Service Models Infrastructure as a Service (IaaS): Key Concepts, Examples, and Providers, Platform as a Service (PaaS): Key Concepts, Examples, and Providers, Software as a Service (SaaS): Key Concepts, Examples, and Providers Comparing IaaS, PaaS, SaaS Cloud Deployment Models Public Cloud: Characteristics and Providers, Private Cloud: Characteristics, Benefits, and Challenges, Hybrid Cloud: Integration of Public and Private Clouds, Community Cloud: Concept and Use Cases	8 Hours

Unit-3	Virtualization in Cloud Computing Basics of Virtualization, Hypervisors: Types and Examples, Virtual Machines (VMs) and Containers, Virtualization Technologies: VMware, Hyper-V, Xen, Docker and Kubernetes Overview	8 Hours
Unit-4	Cloud Storage and Resource Management Cloud Storage Types: Object Storage, Block Storage, File Storage, Distributed File Systems, Cloud Resource Allocation and Load Balancing, Cost and Performance Optimization Techniques, Autoscaling in the Cloud	8 Hours
Unit-5	Cloud Security and Privacy Security Risks and Threats in Cloud Computing, Cloud Security Models and Best Practices, Data Privacy and Regulatory Compliance (GDPR, HIPAA, etc.), Identity and Access Management (IAM), Encryption Techniques and Security Protocols	8 Hours
Textbooks:	1. Cloud Computing: Concepts, Technology & Architecture by Thomas Erl 2. Mastering Cloud Computing by Rajkumar Buyya, Christian Vecchiola, and Thamarai Selvi	
Reference Books:	1. Cloud Computing: Principles and Paradigms by Rajkumar Buyya, James Broberg, Andrzej Goscinski 2. Architecting the Cloud by Michael J. Kavis * Latest editions of all the suggested books are recommended.	
Course Code: BCACI240201	Group G: Cloud Computing and IOT (CI) Discipline Major/Minor Course -15 BCA Semester -II Introduction to Amazon Web Services (AWS)	L- 2 T-0 P-0 C-2
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understand the fundamental concepts of AWS and cloud architecture	
CO-2	Demonstrate proficiency in deploying and managing AWS services like EC2, S3, RDS, and VPC.	
CO-3	Implement AWS security features, including IAM, encryption, and monitoring tools.	
CO-4	Utilize cost management tools to optimize AWS services and monitor expenses.	
CO-5	Apply AWS solutions to real-world business problems and scenarios.	
Course Content		
Unit-1	Introduction to AWS and Cloud Computing Overview of Cloud Computing, AWS Cloud Architecture, AWS Global Infrastructure: Regions, Availability Zones, Edge Locations, Benefits and Use Cases of AWS.	8 Hours
Unit-2	AWS Compute Services, AWS Storage Services, AWS Networking and Content Delivery, AWS Databases	8 Hours
Unit-3	AWS Security and Identity Identity and Access Management (IAM), AWS Security Groups, NACLs, and Encryption, AWS Key Management Service (KMS) and CloudHSM	8 Hours
Unit-4	AWS Cost Management and Monitoring AWS Pricing Models: On-Demand, Reserved, and Spot Instances, AWS Cost Explorer and Budgets, AWS Trusted Advisor, AWS CloudWatch for Monitoring Resources	8 Hours
Unit-5	Real-World Applications and Case Studies AWS for E-Commerce, Media, and Healthcare, Cloud Migration Strategies using AWS Best Practices for Architecting AWS Solutions Preparing for AWS Certification: AWS Certified Cloud Practitioner Exam Preparation	8 Hours
Textbooks:	1. AWS Certified Solutions Architect Official Study Guide by Joe Baron, Hisham Baz, and Tim Bixler 2. AWS: The Complete Guide from Beginner to Advanced by Dennis Hutten	
Reference Books:	1. Amazon Web Services in Action by Andreas Wittig and Michael Wittig	

	* Latest editions of all the suggested books are recommended.	
Group H: Digital Marketing (DM)		
Course Code: BCADM240101	Group H: Digital Marketing (DM) Discipline Major/Minor Course -16 BCA Semester -I Basics of Digital Marketing	L- 4 T-0 P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understand the basic concepts and principles of digital marketing.	
CO-2	Identify and utilize various digital marketing channels and strategies, including SEO, social media, and email marketing.	
CO-3	Create and optimize content for digital marketing purposes.	
CO-4	Develop and implement digital marketing campaigns across multiple platforms.	
CO-5	Analyze and measure the performance of digital marketing campaigns using key metrics and tools.	
Course Content		
Unit-1	Introduction to Digital Marketing Overview of Marketing in the Digital Age, Traditional vs. Digital Marketing, Key Components of Digital Marketing, Importance of Digital Marketing for Businesses, Digital Marketing Terminology.	8 Hours
Unit-2	Search Engine Optimization (SEO), Content Marketing, Social Media Marketing (SMM), Email Marketing	8 Hours
Unit-3	Pay-Per-Click (PPC) and Paid Advertising: Introduction to Google Ads and Bing Ads, Keyword Planning and Ad Copywriting, Managing PPC Campaigns, Retargeting and Display Ads, Budgeting and Bid Strategies	8 Hours
Unit-4	Analytics and Data-Driven Marketing: Introduction to Google Analytics, Tracking Website Traffic and Conversions, Setting Goals and KPIs, A/B Testing and Experimentation, Using Data to Optimize Campaigns	8 Hours
Unit-5	Digital Marketing Strategy and Case Studies Building a Digital Marketing Strategy, Case Studies: Successful Digital Marketing Campaigns, Applying Digital Marketing to Different Industries, Best Practices for Developing and Executing Campaigns	8 Hours
Textbooks:	1. Digital Marketing for Dummies by Ryan Deiss and Russ Henneberry 2. SEO 2024: Learn Search Engine Optimization with Smart Internet Marketing Strategies by Adam Clarke	
Reference Books:	1. Content Marketing: How to Build a Great Brand and Gain High-Quality Customers by John Gibb and Anette Curzon 2. The Art of Digital Marketing: The Definitive Guide to Creating Strategic, Targeted, and Measurable Online Campaigns by Ian Dodson * Latest editions of all the suggested books are recommended.	
Course Code: BCADM240201	Group H: Digital Marketing (DM) Discipline Major/Minor Course -17 BCA Semester -II Web Fundamentals	L- 4 T-0 P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understanding the basics of web technologies, HTML, Linking of HTML files.	
CO-2	Understanding the concept of Image alignment text alignment in HTML.	
CO-3	Understanding various operations on cascading style sheet (CSS).	
CO-4	Understanding the bootstrap framework with its features and layout.	

CO-5	Understanding the concept of web page working, creating forms using of buttons.	
Course Content		
Unit-1	Basics of Web-Technology: HTML – URI, LIST, Hyperlinks: History of HTML, Introduction to URI: Fragment Identifier & Relative Uniform Resource indicator, Standard Generalized Markup Language, Structure of HTML document, Switching between your Editor and Browser, Structuring Web Page, Paragraph and Line Break Tags, Adding Comments, Formatting your Text; Creating Lists: Ordered List Tags, Unordered List Tag & Nesting Lists: Controlling How Ordered Lists are displayed, Creating a Multilevel Outline, Using Start and Value Attributes in an Ordered List, Controlling the Display of Unordered List, Creating Definition List; Creating Hyper Text Links, Linking to a File or Data Object, Linking to NON-WWW Files, Linking to a Place in the Same HTML File, Linking to a Place in Another HTML File, Creating Link Lists, Creating a Simple Link List.	8 Hours
Unit-2	HTML – Images, Links, Rules, Address Tag and Text: Inserting Images: Using the Align Attribute in Inline Graphics, Setting the Height and Width of an Inline Image; Creating Image Links; Horizontal Rules: Changing the Height of a Horizontal Rule, Changing between Shaded and Un-shaded Horizontal Rule, Changing the Width of a Horizontal Rule, Setting the Alignment of a Horizontal Rule; Address Tag; Working with Text: Text Alignment, Changing Font Sizes and Colors: Setting Font Sizes, Setting the Base Font, Using the Small and Big tags, Changing the Font Color; Using a Background Image; Marquee Tag	8 Hours
Unit-3	Web Page Authoring using HTML: Tables: Creating Columns and Rows, Adding a Border, Adding Column Headings, Adding Spacing and Padding, Adding a Caption, Setting the Table Width and Height, Aligning Cell Contents, Setting Column Width, Centering a Table, Inserting an Image, Spanning Columns & Spanning Rows, Setting Font Size and Colors, Assigning Background Colors; Frames: Percentage dimensions, Relative dimensions, Creating two rows Frames, Creating two columns frames, Creating two rows and the second row containing two columns; Forms: What is Form?, Form Tag, Method, Action, Input Tag, Type Attribute: Check box, Hidden, Image, Radio, Reset, Submit, Text Cascading Style Sheets (CSS): Properties Table: Using the style Attribute, Creating Classes and IDs, Generating External Style Sheets, Typography, Consistency, Types of styles, Specifying class within HTML document, Style placement: Inline style, Span & div tags, header styles, Text and font attributes: Font Vs CSS, changing fonts, text attributes, Advance CSS properties: Backgrounds, Box properties and Positioning	8 Hours
Unit-4	Bootstrap Framework, History of Bootstrap, Advantages of Bootstrap Framework, What is Responsive web page, Major Features of Bootstrap, What is Mobile-First Strategy, How to apply Bootstrap to Applications, Bootstrap Grid, Advantages of Bootstrap Grid, Display responsive Images, use readymade themes. Bootstrap Tables, Bootstrap Form Layout.	8 Hours
Unit-5	HTML – Designing static HTML Pages using tags: Textbox, Button, Radio Button, Check Box, Text Area, Image, Links, Anchors, Table, Lists, Dropdown List; Form Submission using Get and Post Methods; JavaScript: Adding JavaScript to static HTML pages; Publishing a website.	8 Hours
Textbooks:	1. Burdman, Collaborative Web Development , Addison Wesley.	
Reference Books:	1. HTML & CSS: Design and Build Websites, Jon Duckett * Latest editions of all the suggested books are recommended.	
Group I: Cloud Technology and Information Security (CTIS)		
Course Code: BCACT240101	Group I: Cloud Technology and Information Security (CTIS) Discipline Major/Minor Course -18 BCA Semester -I	L- 4 T-0 P-0

	Introduction to Cloud Technology	C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Explain the core issues of cloud computing such as security, privacy, and interoperability.	
CO-2	Understand the concepts, characteristics, delivery models and benefits of cloud computing.	
CO-3	Examine the appropriate technologies, algorithms, and approaches for the related issues.	
CO-4	Identify problems, and explain, analyze, and evaluate various cloud computing solutions	
CO-5	Identify the architecture and infrastructure of cloud computing, including SaaS, PaaS, IaaS, public cloud, private cloud, hybrid cloud, etc.	
Course Content		
Unit-1	Introduction to Cloud Computing Introduction to Cloud Computing, History and Evolution of Cloud Computing, Types of clouds, Private Public and hybrid clouds, Cloud Computing architecture, Cloud computing infrastructure, Merits of Cloud computing, , Cloud computing delivery models and services (IaaS, PaaS, SaaS), obstacles for cloud technology, Cloud vulnerabilities, Cloud challenges, Practical applications of cloud computing.	8 Hours
Unit-2	Cloud Computing Companies and Migrating to Cloud Web-based business services, Delivering Business Processes from the Cloud: Business process examples, Broad Approaches to Migrating into the Cloud, The Seven-Step Model of Migration into a Cloud, Efficient Steps for migrating to cloud., Risks: Measuring and assessment of risks, Company concerns Risk Mitigation methodology for Cloud computing, Case Studies	8 Hours
Unit-3	Cloud Cost Management and Selection of Cloud Provider Assessing the Cloud: software Evaluation, System Testing, Seasonal or peak loading, Cost cutting and cost-benefit analysis, Selecting the right scalable application. Considerations for selecting cloud solution. Understanding Best Practices used in selection of Cloud service and providers, Clouding the Standards and Best Practices Issue: Interoperability, Portability, Integration, Security, Standards Organisations and Groups associated with Cloud Computing, Commercial and Business Consideration	8 Hours
Unit-4	Governance in the Cloud Industry Standards Organizations and Groups associated with Cloud Computing, Need for IT governance in cloud computing, Cloud Governance Solution: Access Controls, Financial Controls, Key Management and Encryption, Logging and Auditing, API integration. Legal Issues: Data Privacy and Security Issues, Cloud Contracting models, Jurisdictional Issues Raised by Virtualization and Data Location, Legal issues in Commercial and Business Considerations	8 Hours
Unit-5	Ten Cloud do's and do not's Don't be reactive, do consider the cloud a financial issue, don't go alone, do think about your architecture, don't neglect governance, don't forget about business purpose, do make security the centerpiece of your strategy, don't apply the cloud to everything, don't forget about Service Management, do start with a pilot project	8 Hours

Textbooks:	1. <i>Cloud Computing: Principles and Paradigms</i> , Rajkumar Buyya, James Broberg, Andrzej M. Goscinski, John Wiley and Sons Publications, 2011 2. <i>Brief Guide to Cloud Computing</i> , Christopher Barnett, Constable & Robinson Limited, 2010	
Reference Books:	1. <i>Handbook on Cloud Computing</i> , Borivoje Furht, , Armando Escalante, Springer, 2010 * <i>Latest editions of all the suggested books are recommended.</i>	
Course Code: BCACT240201	Group I: Cloud Technology and Information Security (CTIS) Discipline Major/Minor Course -19 BCA Semester -II Linux Administration	L- 4 T-0 P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understanding how to manage and administer systems and servers using Linux Operating System.	
CO-2	Understanding the operating system used for this unit is Red hat Enterprise Linux 6 (RHEL 6).	
CO-3	Understanding the local storage and file system.	
CO-4	Applying highly scalable MySQL or Oracle database in the cloud with multiple read-replica databases (for scalability of database)	
CO-5	Applying setup a load-balancer in the cloud.	
Course Content		
Unit-1	Introduction to Linux Introduction to Operating system - Types of Operating system - Multi user operating system - Open source licensing - History of Linux - Unix Vs Linux - Flavors of Linux - Benefits and characteristics of Linux - Installation of Linux - Linux booting process - Log in and switch users in multiuser run levels - Shell and bash features - Linux kernel - sudo/su - Date and time configuration – Linux run levels. Directories and files: Directory structure - System directory - Absolute path and relative path - Creating and removing directory - Changing directory path - Creating - removing - copying and moving files - File Permissions - Links – hard link and soft link - Input and output redirection - Filters and pipes - Locate - read - and use system documentation including man page	8 Hours
Unit-2	Package, User and group Management RPM - YUM - Archive - Compress - unpack and decompress files using tar - star - gzip - and bzip2 - Create - delete - and modify local user accounts - Change passwords for local user accounts - Create - delete - and modify local groups and group memberships - Changing owner and modes.	8 Hours
Unit-3	Configuring local storage and file system List - create - delete - and partition type for primary - extended - and logical partitions - Create and remove physical volumes - assign physical volumes to volume groups - Create and delete logical Volumes. - Create - mount – un mount - ext2 - ext3 - and ext4 file systems. - Mount – un mount-and LUKS-encrypted file systems - Access control list.	8 Hours
Unit-4	Managing system and infrastructure services: Managing system services - Shutting down - suspending and hibernating the system - Controlling system on remote machine - Creating and modifying system unit files – DHCP Configuration - HTTP server Configuration - FTP	8 Hours

	server Configuration - Mail server Configuration - Samba server Configuration - NTP server Configuration - NFS server Configuration	
Unit-5	Open SSH and Linux security: OPENSsh - The SSH Protocol - Configuring Open SSH and Starting an Open SSH Server Key-Based Authentication in Open SSH - Open SSH Clients - Using the ssh Utility - scp Utility and sftp Utility - Configure firewall settings using system-config-firewall or iptables - Set enforcing and permissive modes for SE Linux - List and identify SE Linux file and process context.	8 Hours
Textbooks:	1. Orsaria, Jang, “RHCSA/RHCE Red Hat Linux Certification Study Guide Exams EX200 & EX300”, McGraw-Hill Education, July 2017.	
Reference Books:	1. Sander Van Vugt, “Red Hat RHCSA/RHCE 7 Cert Guide: Red Hat Enterprise Linux 7 (EX200 and EX300)”, Phi Learning Pvt Ltd, 2009. <i>*Latest editions of all the suggested books are recommended.</i>	
Group J: Mobile Application and Web Technology (MAWT)		
Course Code: BCAMW240101	Mobile Application and Web Technology (MAWT) Discipline Major/Minor Course -20 BCA Semester -I Introduction to Mobile Device and Network Architecture	L- 4 T-0 P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Understand the fundamental principles of wireless communication.	
CO-2	Learn the foundational concepts of cellular networks.	
CO-3	Explore the GSM protocol stack.	
CO-4	Gain knowledge of CPU, memory, and power management in handsets.	
CO-5	Explain the evolution of hardware architecture.	
Course Content		
Unit-1	Communication Principles Wireless Communication Principles, Radio Communication, Analogy and Digital Communication, Benefits of Digital Signals, Computer Network, OSI Model, Mobile Networks ,Mobile Network Protocol , Telecommunication Networks, Fixed Networks,	8 Hours
Unit-2	Mobile Cellular Networks Cellular Network Concepts, Cells and Base Stations, Mobile Network Architecture, Mobile Network, Mobile Network Subsystems, Mobile Network Protocol Stacks, PLMN (Public Land Mobile Network), Mobile Network Fundamentals, Mobile Network Features, Mobility, Registration, Handoff, Roaming.	8 Hours
Unit-3	GSM and CDMA Networks: GSM History, GSM RF Channels, 2G Network Architecture, GSM Protocol Stack, GPRS Standards, CS and PS Domains, GPRS Architecture, GPRS Protocols, CDMA Evolution, 2G CDMAOne, CDMA 2G Standards, 3GPP2 Network Architecture, Mobile IP, UMTS Spectrum, UMTS Radio Access Network, UMTS Protocol Stacks, SIP Network, UMTS Multiple Access Network Architecture, 4G	8 Hours
Unit-4	Handset Evolution, Handset Characteristics and Features Mobile Phone and Network Evolution, Mobile Phones, Mobile Handset Characteristics. Wireless Cellular. Bluetooth . Display. Keypad. Camera. Mobile	8 Hours

	Handset Categories, Low end Phones, Feature Phones, Smart phones, Handset Components, Handset Design, Handset Manufacture, Handset Bill of Materials, assembling handsets.	
Unit-5	Hardware Architecture and Subsystems Handset hardware architecture, Primary Hardware Subsystems, Element inside a Mobile Handset, Hardware architectural trends, CPU and Memory, Hardware evolution, Introduction to the Radio subsystems, Function of the RF Subsystems, Handset Power Requirements, Power Management, Introduction and Definition to the SIM, Smartcards in general and concept of Mobile Identity , Phones without SIMs,	8 Hours
Textbooks:	1. Wireless and Mobile Network Architectures by Yi-Bang Lin and Imrich Chlamtac, Wiley-India, 2008 2. Mobile Networks Architecture by <u>Andre Perez</u> , Wiley, March 2012	
Reference Books:	1. Mobile Computing – Technology, Application & Service Creation by Asoke. K Talukder, Roopa R. Yavagal, Asoke K. Talukder, Tata McGraw-Hill, 2005 2. GSM - Architecture, Protocols and Services by <u>Jörg Eberspächer</u> , <u>Hans-JoergVögel</u> , <u>Christian Bettstetter</u> , <u>Christian Hartmann</u> John Wiley & Sons, Dec-2008 * Latest editions of all the suggested books are recommended.	
Course Code: BCAMW240201	Mobile Application and Web Technology (MAWT) Discipline Major/Minor Course -21 BCA Semester -II User Interface and User Experience	L- 4 T-0 P-0 C-4
Course Outcomes:	On completion of the course, the students will be:	
CO -1	Through this course students will be able to learn how the users interact with the computers.	
CO-2	They will get to know the different types of models, design process and interaction styles to develop a good user interface.	
CO-3	They will also get to know different types of designing tools for UX designers	
CO-4	Explore error prevention strategies such as interlock, lock-in, lockout, and behavior shaping constraints.	
CO-5	Learn guidelines for error recovery and the effective use of error messages in UI design.	
Course Content		
Unit-1	INTRODUCTION TO HCI The human –Input Output Channels, Human Memory, Thinking: Reasoning and Problem Solving, The Computer – Text Entry Devices, Positioning, Pointing and Drawing, Display Devices, Memory, The Interaction – Models of Interaction, Frameworks & HCI. Cognitive Models, Communication & Collaboration Models, Task Analysis, Dialog Analysis & Design, Interaction Models, Programming Interactive System.	8 Hours
Unit-2	USER INTERFACE DESIGN The Software Life Cycle – Activities in the Life Cycle, Validation & Verification, Interactive Systems & Software Life Cycle, Using Design Rules – Standards, Guidelines. Usability Engineering – Problems with usability engineering. Iterative	8 Hours

	Design and Prototyping, Design Rationale – Process-oriented design rationale, Design space analysis, Psychological design rationale. Direct Manipulation and Virtual Environments, Menu Selection Form Filling and Dialog Boxes, Command and Natural Languages, Interaction Devices, Collaboration and Social Media Participation, Naming and Abbreviations.	
Unit-3	MOBILE UI DESIGN Non Perceptual Technologies – Keypads, Touchpad, Gesture: Classification of Gesture based HCI – Application Domains: Multimodal interactions, Desktop, Virtual and Augmented Reality – Communication, Gesture Styles – Deictic, Gesticulation, Manipulation, Semaphores, Sign Language, Disruptive Innovation. UX Tools for Wire framing and Prototyping – UXPin, Wireframe.cc, Pixate, Sketch, Invision, P.O.P. app, UX Tools for User Research and User Testing – Lookback, Wufoo, Typeform, Usabilla, Morae, UX, Recorder, UX Tools for Organizing Information – Lucid Chart, MindMeister, Coggle.it, Gliffy.	8 Hours
Unit-4	BEST PRACTICES IN UI DESIGN Views and Layout Tools – Paragraph Tag, Image Tag, Table Tag, Layout Tag, Frame Tag, Interaction – Links, Buttons, Text Fields, Radio Buttons, Check Boxes, CSS – Syntax, Colors and Background, Text and Fonts, Icons and Links, List. JavaScript – Syntax, Statements, Variables, Data Types, Operators, Events, JS Forms – Forms API, JS Objects – Object Definition, Object Properties, Object Methods, Object Prototypes, JS Functions – Function Definition, Function Parameters, Function Invocation, Function Closures.	8 Hours
Unit-5	CASE STUDIES Universal Usability: Web Fun for Individuals with Down syndrome, Blind Users Experience The Internet, Using The “Technology as Experience” Framework, Establishing Requirements for a Mobile Learning System, Using XP to Develop Context-Sensitive Adverts for The Web, Designing Mobile Applications for Multiple Form Factors. Types of Errors – Mistakes, Slips, False Understanding, Slips – Capture Error, Description Error, Loss of Activation, Mode Errors, Strategies : Prevention – Interlock, Lock in, Lockout, Behavior Shaping Constraints, Error Recovery – Guidelines, Error Messages.	8 Hours
Textbooks:	1. “Human Computer Interaction” by Alan Dix, Janet Finlay, ISBN: 9788131717035, Pearson Education (2004) 2. “Designing the User Interface - Strategies for Effective Human Computer Interaction”, by Ben Shneiderman. Pearson Education (2010). 3. The Essential Guide to User Interface Design: An Introduction to GUI Design Principles and Techniques, by Wilbert O. Galitz. Wiley (2007)	
Reference Books:	1. Usability Engineering: Scenario-Based Development of Human-Computer Interaction , by Rosson, M. and Carroll, J. (2002) 2. The Essentials of Interaction Design, by Cooper, et al. , Wiley Publishing(2007) 3. Usability Engineering, by Nielsen, J. Morgan Kaufmann, San Francisco, 1993. ISBN 0-12-518406-9 4. The Resonant Interface: HCI Foundations for Interaction Design , by Heim, S. , Addison-Wesley. (2007) 5. Usability engineering: scenario-based development of human-computer interaction, By Rosson, M.B & Carroll. J.M. , Morgan Kaufman.(2002). * Latest editions of all the suggested books are recommended.	
Ability Enhancement Course (AEC)		
Course Code TMUAE240151	Ability Enhancement Course -1	L-1 T-0

	BCA Semester -I English Communication	P-2 C-2
Course Outcomes:	On completion of the course, the student will be:	
CO1.	Conversing confidently in everyday English, using broad vocabulary andproper sentence structures.	
CO2.	Participating effectively in diverse discussions and expressing opinionswith proficiency in complex language functions.	
CO3.	Using a range of grammatical structures intuitively for fluent, accuratecommunication.	
CO4.	Applying self-assessment strategies for continuous language improvementand effective use of English.	
CO5.	Demonstrating English usage in both formal and informal contexts throughpractical experience.	
Course Content		
Unit 1:	Introduction to Spoken English (Laying Foundations) - Module 1: Greetings and Introduction - Module 2: Expressing likes and dislikes - Module 3: Talking about family and friends - Module 4: Discussing daily routines - Module 5: Describing physical environment	09 hours
Unit 2:	Engaging in Conversations - Module 1: Asking and answering simple questions - Module 2: Food and eating out - Module 3: Shopping and transactions - Module 4: Talking about hobbies and interests - Module 5: Describing health and symptoms	10 hours
Unit 3:	Practical Language Application - Module 1: Discussing past experiences - Module 2: Making future plans - Module 3: Giving and following directions - Module 4: Expressing opinions and preferences - Module 5: Handling emergencies and unexpectedsituations	10 hours
Unit 4:	Advanced Conversation Skills - Module 1: Discussing career and education - Module 2: Cultural awareness and global issues - Module 3: Technology and modern life - Module 4: Environment and sustainability - Module 5: Arts and entertainment	10 hours
Unit 5:	Mastery and fluency development	09 hours

	- Module 1: Review and consolidation - Module 2: Advanced dialogue and conversation strategies - Module 3: Public speaking and presentation skills - Module 4: Small talks and advancing conversations - Module 5: Communicating spontaneously and unprepared	
Methodology	1. Every module will be composed of two sessions namely Prep (Preparation) and Convo (Conversation) sessions. 2. Sessions will be 100% activity and participation based. 3. Handouts and manuals will be delivered to the students in each session and will supplement the training sessions. 4. Every handout will consist of 3 essential sections namely Key vocabulary, Key sentence structure and Supporting Activities. 5. Accountability partners will be assigned to each participant for a sense of mutual responsibility towards achievements of each other's learning outcomes. The final internal evaluation will be in terms of an engaging competition rather than a conventional exam.	
Textbooks	1. McCarthy, M., O'Dell, F., Redman, S. <i>English Vocabulary in Use Elementary/Pre-intermediate & Intermediate</i>. Cambridge University Press 2. McCarthy, M., O'Dell, F., Redman, S. <i>English Phrasal Verbs in Use Intermediate</i>. Cambridge University Press 3. Yates, J. (2012) <i>English Conversation. Practice makes Perfect Series</i>, Mc Graw Hill	
Reference Books	1. Murphy, R. <i>English Grammar in Use</i>. Cambridge University Press 2. Swan, M. <i>Practical English Usage</i>. Oxford University Press * Latest editions of all the suggested books are recommended.	
Course Code TMUAE240551	Ability Enhancement Course-2 BCA Semester -V Personal Effectiveness	L-2 T-0 P-0 C-2
Course Outcomes:	On completion of the course, the student will be:	
CO1.	Demonstrating foundational soft skills necessary for personal and Professional growth.	
CO2.	Applying principles of emotional intelligence for effective self-management and interpersonal relationships.	
CO3.	Analyzing and synthesizing information to solve complex problems efficiently.	
CO4.	Participating effectively in group discussions, including communication, collaboration, and critical thinking.	
CO5.	Developing the skills and strategies needed to excel in job interviews and building a compelling resume.	
Course Content		
Unit 1:	General Soft Skills - Introduction to Communication Skills - Right Attitude and Self-Motivation - Perception	6 hours

	• Time Management • Stress Management	
Unit 2:	Emotional Intelligence • Introduction to emotional intelligence • Self-awareness and self-regulation • Empathy and social skills • Managing emotions in various situations	6 hours
Unit 3:	Problem-Solving and Decision-Making • Problem identification and analysis • Creative thinking techniques • Decision-making models and strategies • Implementing effective solutions	6 hours
Unit 4:	Understanding Group Discussion • Introduction to Group Discussion (GD) and its purpose in recruitment processes • Components of a successful GD: communication, collaboration, and critical thinking • Types of Topics in Group Discussion: Political, Economical, Social, Technological, Abstract • Identifying common challenges in GDs and strategies to overcome them • Practicing active listening and respectful participation in group settings	8 hours
Unit 5:	Interview Skills and Resume Building • Resume and Cover Letter writing • Interview preparation strategies • Handling different interview formats • Common Interview Questions • Mock interviews and feedback sessions	10 hours
Textbooks	1. Robbins, S. P., & Judge, T. A. (2018). <i>Organizational Behavior</i> (18th ed.). Pearson. 2. Goleman, D. (2006). <i>Emotional Intelligence: Why It Can Matter More Than IQ</i>. Bantam. 3. Dyer, J. H., Gregersen, H. B., & Christensen, C. M. (2009). <i>The Innovator's DNA: Mastering the Five Skills of Disruptive Innovators</i>. Harvard Business Review Press.	
Reference Books	1. Covey, S. R. (2004). <i>The 7 Habits of Highly Effective People: Powerful Lessons in Personal Change</i> (25th Anniversary ed.). Free Press. 2. Tuckman, B. W. (1965). Developmental Sequence in Small Groups. <i>Psychological Bulletin</i>, 63(6), 384–399. https://doi.org/10.1037/h0022100 3. Cialdini, R. B. (2006). <i>Influence: The Psychology of Persuasion</i> (Rev. ed.). Collins. * Latest editions of all the suggested books are recommended.	
Course Code TMUAE240601	Ability Enhancement Course-3 BCA Semester -VI	L-2 T-0 P-0

	Team Building and Leadership	C-2
Course Outcomes:	On completion of the course, the student will be:	
CO1.	Understanding significance, relevance of teams' development and function of effective teams, contributing to team cohesiveness.	
CO2.	Utilizing Tuckman's model of team building based on characteristics of performing organization, teams and team members	
CO3.	Functioning as leader of small team by utilizing communication skills and achieving trust and collaboration with proper understanding of concept of Leader and manager using knowledge of management theories.	
CO4.	Leading a team to create proper vision, mission and goals by empowering and motivating team members.	
CO5.	Developing strategies for achieving set goals and coaching and mentoring team members to achieve set goals	
Course Content		
Unit 1:	Module 1.1: Understanding Team Dynamics - Outcomes of Team Building and Teamwork - Philosophy and significance of teamwork - Why working in teams is even more relevant today - Characteristics of performing organizations, effective teams and team members - Team Building stages (Tuckman's model: Forming, Storming, Norming, Performing, Adjourning) - Role clarification and team member expectations	05 hours
	Module 1.2: Enhancing Team Effectiveness - Communication skills within teams - Conflict resolution strategies - Building trust and cohesion, Cooperation / collaboration - Decision-making processes within teams	05 hours
Unit 2:	Module 2.1: Introduction to Leadership - Definition and importance of right leadership - Functions of a leader - Differentiating between leadership and management - Historical perspectives on leadership theories Contemporary approaches to leadership studies	04 hours
	Module 2.2: Theories and Styles of Leadership - Trait theory: Identifying key leadership traits - Behavioral theory: Understanding behavioral approaches to leadership - Leadership Styles: Power Style, Orientation Style, Transformation Style, Transactional Style, Charismatic Leadership Style Ideal leader inner hard core.	04 hours
Unit 3:	Module 3.1: Role of Leadership in Team Success - Importance of leadership in team effectiveness - Creating a shared vision and goals for the team	04 hours

	- Facilitating effective communication within the team - Motivating and empowering team members	
	Module 3.2: Leading High-Performing Teams - Strategies for building and maintaining high-performing teams - Why there is scarcity of good leaders - Coaching and mentoring team members - Delegating tasks effectively - Monitoring team progress and providing feedback	04 hours
Unit 4:	Module 4.1: Case Studies in Teamwork - Analysis of real-world team dynamics and challenges - Identification of successful teamwork strategies - Lessons learned from failed team endeavors	04 hours
	Module 4.2: Leadership Development Activities - Leadership simulations and role-playing exercises - Reflection on personal leadership styles and strengths - Peer feedback sessions for leadership improvement - Creating a personal leadership development plan	04 hours
Assessments Methodology	- Written assignments analyzing teamwork dynamics and leadership theories - Group presentations on case studies with teamwork and leadership insights - Participation in team-based activities and reflection on leadership experiences	
Textbooks	- Northouse, P. G. (2021). <i>Leadership: Theory and practice</i> (9th ed.). SAGE Publications. - Yukl, G. (2021). <i>Leadership in organizations</i> (10th ed.). Pearson. - Kouzes, J. M., & Posner, B. Z. (2017). <i>The leadership challenge: How to make extraordinary things happen in organizations</i> (6th ed.). Jossey-Bass.	
Reference Books	- Williams, R., & Weber, K. (2019). <i>Learning to lead: The journey to leading yourself, leading others, and leading an organization</i>. Greenleaf Book Group Press. - Maxwell, J. C. (2019). <i>Leadership: The 11 essential changes every leader must embrace</i>. HarperCollins Leadership. - Maxwell, J. C. (2018). <i>Developing the leader within you 2.0</i>. HarperCollins Leadership. - Schein, E. H., & Schein, P. A. (2018). <i>Humble leadership: The power of relationships, openness, and trust</i>. Berrett-Koehler Publishers. * Latest editions of all the suggested books are recommended.	
Course Code TMUAE240701	Ability Enhancement Course -4 BCA Semester -VII Corporate Skills	
		L- 1 T-0 P-2 C-2
Course Outcomes:	On completion of the course, the student will be:	
CO1.	Applying team-building, trust-building, and collaboration skills to enhance Influence within teams.	

CO2.	Implementing professional branding, business etiquette, goal-setting, and action planning for personal and career development.	
CO3.	Demonstrating cultural awareness, adapting communication styles, promoting workplace sensitization, and delivering effective presentations.	
CO4.	Utilizing advanced techniques for leading group discussions, managing diverse opinions, and practicing different GD formats for effective communication.	
CO5.	Applying advanced interview strategies including company research, Utilizing behavioral interview techniques, demonstrating professional demeanor, and using non-verbal communication for successful interviews.	
Course Content		
Unit 1:	Teamwork and Influence • Team building • Building trust and credibility • Networking and collaboration	6 hours
Unit 2:	Professional Growth • Professional Branding and Image Building • Business etiquette and protocol • Setting SMART goals Developing action plans for success	6 hours
Unit 3:	Corporate Communication • Cultural awareness and sensitivity • Different Communication styles • Workplace sensitization Presentation Skills	6 hours
Unit 4:	Excelling in Group Discussions • Advanced techniques for leading group discussions • Different GD formats: topic-based, case study-based, and open-ended discussions • Managing diverse opinions and perspectives Practice of Group Discussions on common topics	8 hours
Unit 5:	Advanced Interview Strategies • Researching the company • Behavioral interview techniques • Professional attire and etiquette • Body Language and non-verbal communication Real interview stimulations	10 hours
Textbooks	1. Robbins, S. P., & Judge, T. A. (2019). <i>Organizational Behavior</i> (18th ed.). Pearson. 2. Cardon, P. W., & Scott, J. C. (2019). <i>Problem Solving and Decision Making: Illustrated Course Guides</i> (2nd ed.). Cengage Learning. 3. O'Hair, D., Wiemann, M., & Mullin, D. (2015). <i>Real Communication</i> (3rd ed.). Bedford/St. Martin's.	
Reference Books	1. Tuckman, B. W. (1965). <i>Developmental Sequence in Small Groups</i>. <i>Psychological Bulletin</i>, 63(6), 384–399. 2. Covey, S. R. (2004). <i>The 7 Habits of Highly Effective People: Powerful Lessons in Personal Change</i> (25th Anniversary ed.). Free press 3. Cialdini, R. B. (2006). <i>Influence: The Psychology of Persuasion</i> (Rev. ed.). Collins.	

	* Latest editions of all the suggested books are recommended.	
Skill-Enhancement Course		
Course Code TMUSE240101	Skill-Enhancement Course -1 BCA Semester -I Innovation & Entrepreneurship Development	L-3 T-0 P-0 C-3
Course Outcomes:	On completion of the course, the students will be :	
CO1.	Understanding the basic concepts, role and importance of entrepreneurship for the economic development	
CO2.	Understanding the systematic process of selecting and screening of a business idea.	
CO3.	Developing personal creativity and entrepreneurial initiative.	
CO4.	Understanding the concepts and schemes offered by various commercial banks and financial institutions like IDBI, ICICI, SIDBI, SFC. etc.	
CO5.	Analyzing business environment in order to identify business opportunities.	
	Course Content:	
Unit-1:	Introduction to Innovation and Entrepreneurial Idea Generation and Identifying Business Opportunities Concepts of Entrepreneurship Development in Indian Context: Evolution of the concept of Entrepreneur, Entrepreneur Vs. Intrapreneur, Entrepreneur Vs. Entrepreneurship, Entrepreneur Vs. Manager, Attributes and Characteristics of a successful Entrepreneur.	6 Hours
Unit-2:	Creating Entrepreneurial Venture: Business Planning Process, Environmental Analysis - Search and Scanning, Identifying problems and opportunities, Defining Business Idea.	6 Hours
Unit-3:	Project Management: Technical, Financial, Marketing, Personnel and Management Feasibility, Estimating and Financing funds requirement - Schemes offered by various commercial banks and financial institutions like IDBI, ICICI, SIDBI, SFCs.	6 Hours
Unit-4:	Entrepreneur Development Programmes (EDP'S): Types, 7S Model, History of entrepreneurship development in India, Current Status of Entrepreneurship development in India.	6 Hours
Unit-5:	Entrepreneurship Development and Government: Role of Central Government and State Government in promoting Entrepreneurship - Introduction to various incentives, subsidies and grants - Export Oriented Units - Fiscal and Tax concessions available.	6 Hours
Text Books:	1. <i>Entrepreneurship: New Venture Creation - David H. Holt</i>	
Reference Books:	1. <i>Thought Leaders – Shrinivas Pandit</i> 2. <i>Entrepreneurship - Steven Brandt</i> 3. <i>Business Gurus Speak - S.N.Chary</i> 4. <i>Entrepreneurship - Hisrich Peters</i> 5. <i>The Culture of Entrepreneurship - Brigitte Berger</i> 6. <i>Project Management - K. Nagarajan</i> 7. <i>Dynamics of Entrepreneurship Development - Vasant Desai</i> * Latest editions of all the suggested books are recommended.	
Course Code	Skill-Enhancement Course -2	L-3 T-0

TMUSE240201	BCA Semester -II Basic Analytical Skills	P-0 C-3
Course Outcomes:	On completion of the course, the students will be :	
CO1.	Operationalizing the inter-related concept of Percentage in Profit, Loss and Discount.	
CO2.	Applying the arithmetical concepts in Ratio and Proportion, Mixture and Allegation and Average.	
CO3.	Employing the techniques of Percentage, Ratios and Average in inter related concepts of Time and Work, Time speed and Distance.	
CO4.	Evaluating the different possibilities of various reasoning-based problems in series, Blood relation, Direction and Coding-Decoding.	
Course Content:		
	Quantitative Aptitude (QA)	
Unit-1:	Percentages Basic calculation, Ratio equivalent, Multiplying factor, Percentage change, Increment, Decrement, Successive percentages and word problems Profit Loss Discount Basic definition, formula, concept of mark up, discount, relation with successive change, faulty weights	9 Hours
Unit-2:	Ratio, proportions and variations Concept of ratios, proportions, variations, properties and their applications Mixtures and allegations Mixtures of 2 components, mixtures of 3 components, Replacements Averages Basic Averages, Concept of Distribution, Weighted Average, equations	9 Hours
Unit-3:	Time and Work Same efficiency, different efficiency, alternate work, application in Pipes and Cisterns Time Speed Distance Basic concept of Proportionalities, Average speed, Problems based on Trains, Boat and stream, Races- Linear and circular	10 Hours
	Logical Reasoning (LR)	
Unit-4:	Blood Relations Indicating Type, Operator Type (Coded BR), Family Tree Type Direction sense Simple statements, shadow type	4 Hours
Unit-5:	Coding and Decoding Sequential coding, Reverse coding, Abstract coding Number and alphabet series Number series, Alphabet series	4 Hours
Reference Books:	R1: -Arun Sharma: - How to Prepare for Quantitative Aptitude R2: -Quantitative Aptitude by R.S. Agrawal R3: -M Tyra: Quicker Maths R4: -Nishith K Sinha: - Quantitative Aptitude for CAT R5: -Reference website: - Lofoya.com, gmatclub.com, cracku.in, handakafunda.com, tathaqaat.mba, Indiabix.com	

	R6: -Logical Reasoning by Nishith K Sinha R7: -Verbal and Non-Verbal Reasoning by R.S. Agrawal * Latest editions of all the suggested books are recommended.																								
Evaluation Scheme:	➤ 30 marks best 2 out of CT1 + CT2 ➤ 10 marks for attendance. ➤ 60 marks for final external exams. Internal Evaluation = 40 marks (30 marks for CTs + 10 marks for attendance) <table><tr><th colspan="5">Evaluation Scheme – Internal</th></tr><tr><th>Activity</th><th>CT 1</th><th>CT 2</th><th>ASSIGNMENT</th><th>Attendance</th></tr><tr><td>Max Marks</td><td>20</td><td>20</td><td>10</td><td>10</td></tr></table> - Marks of Best CT out of CT 1 & CT 2 will be considered - Each CT would be of 20 marks (total 30 MCQs and 1 hour duration). External = 60 marks <table><tr><th colspan="2">Evaluation Scheme - External</th></tr><tr><th>Activity</th><th>MCQ based Test on OMR</th></tr><tr><td>Max Marks</td><td>60</td></tr></table> - Single Paper of 60 Questions, each carrying 1 mark and 90 min (1 and half hour) duration.				Evaluation Scheme – Internal					Activity	CT 1	CT 2	ASSIGNMENT	Attendance	Max Marks	20	20	10	10	Evaluation Scheme - External		Activity	MCQ based Test on OMR	Max Marks	60
	Evaluation Scheme – Internal																								
	Activity	CT 1	CT 2	ASSIGNMENT	Attendance																				
	Max Marks	20	20	10	10																				
	Evaluation Scheme - External																								
Activity	MCQ based Test on OMR																								
Max Marks	60																								
Course Code TMUSE240301	Skill-Enhancement Course -3 BCA Semester -III Advance Analytical Skills			L-3 T-0 P-0 C-3																					
Course Outcomes:	On completion of the course, the students will be :																								
CO1.	Applying the concepts of modern mathematics Divisibility rule, Remainder Theorem, HCF /LCM in Number System.																								
CO2.	Relating the rules of permutation and combination, Fundamental Principle of Counting to find the probability.																								
CO3.	Applying calculative and arithmetical concepts of ratio, Average and Percentage to analyze and interpret data.																								
CO4.	Employing the concept of higher-level reasoning in Clock and Calendars, Syllogism and Puzzle Problems.																								
Course Content:																									
Quantitative Aptitude (QA)																									
Unit-1:	Number theory Classification of Numbers, Divisibility Rules, HCF and LCM, Factors, Cyclicity (Unit Digit and Last Two digit), Remainder Theorem, Highest Power of a Number in a Factorial, Number of trailing zeroes SI and CI Simple Interest, finding time and rate, Compound Interest, difference between SI			8 Hours																					

	and CI, Concept of Installments																
Unit-2:	Permutations and combinations Fundamental counting, and or, arrangements of digits, letters, people in row, identical objects, rank, geometrical arrangements, combination: basic, handshakes, committee, selection of any number of objects, identical and distinct, grouping and distribution, de-arrangements Probability Introduction, Probability based on Dice and Coins, Conditional Probability, Bayes Theorem	8 Hours															
Data interpretation and Data Sufficiency (DI & DS)																	
Unit-3:	Data Interpretation Basics, Bar Chart, Line Chart, Tabular Chart, Pie Chart, DI tables with missing values Introduction of Data Sufficiency, different topics-based DS	8 Hours															
Logical Reasoning (LR)																	
Unit-4:	Clocks and calendars Introduction, Angle between hands, Gain and loss of Clock, Interchange of hands, Introduction of Calendars, Leap Year, Ordinary Year	8 Hours															
Unit-5:	Syllogisms Two statements, three statements Problem Solving Introduction, Puzzle based on 3 variables, Puzzle based on 4 variables	10 Hours															
Reference Books:	R1: -Arun Sharma: - How to Prepare for Quantitative Aptitude R2: -Quantitative Aptitude by R.S. Agrawal R3: -M Tyra: Quicker Maths R4: -Nishith K Sinha: - Quantitative Aptitude for CAT R5: -Reference website: - Lofoya.com, gmatclub.com, cracku.in, handakafunda.com, tathagat.mba, Indiabix.com R6: -Logical Reasoning by Nishith K Sinha R7: -Verbal and Non-Verbal Reasoning by R.S. Agrawal * Latest editions of all the suggested books are recommended.																
Evaluation Scheme:	➤ 30 marks best 2 out of CT1 + CT2 + CT3 ➤ 10 marks for attendance. ➤ 60 marks for final external exams. Internal Evaluation = 40 marks (30 marks for CTs + 10 marks for attendance) <table><tr><th colspan="5">Evaluation Scheme – Internal</th></tr><tr><th>Activity</th><th>CT 1</th><th>CT 2</th><th>Assignment</th><th>Attendance</th></tr><tr><td>Max Marks</td><td>20</td><td>20</td><td>10</td><td>10</td></tr></table> • Marks of Best out of 2 CTs will be considered out of CT 1& CT 2.		Evaluation Scheme – Internal					Activity	CT 1	CT 2	Assignment	Attendance	Max Marks	20	20	10	10
Evaluation Scheme – Internal																	
Activity	CT 1	CT 2	Assignment	Attendance													
Max Marks	20	20	10	10													

	- Each CT would be of 15 marks (total 30 MCQs and 1 hour duration). External = 60 marks <table border="1"><thead><tr><th colspan="2">Evaluation Scheme - External</th></tr></thead><tbody><tr><td>Activity</td><td>MCQ based Test on OMR</td></tr><tr><td>Max Marks</td><td>60</td></tr></tbody></table> - Single Paper of 60 Questions, each carrying 1 mark and 90 min (1 and half hour) duration.	Evaluation Scheme - External		Activity	MCQ based Test on OMR	Max Marks	60
Evaluation Scheme - External							
Activity	MCQ based Test on OMR						
Max Marks	60						
Course Code TMUSE240401	Skill-Enhancement Course -4 BCA Semester -IV Modern Office Management	L-3 T-0 P-0 C-3					
Course Outcomes:	On completion of the course, the students will be :						
CO1.	Identifying and describing the fundamental principles and practices of modern office management. <i>(Remembering, Understanding)</i>						
CO2.	Demonstrating effective communication and coordination skills within an office environment. <i>(Applying)</i>						
CO3.	Utilizing modern technology and office automation tools to enhance office efficiency. <i>(Applying)</i>						
CO4	Analyzing and implementing records management and information retrieval systems. <i>(Analyzing, Applying)</i>						
CO5.	Evaluating office layout and ergonomics to design a productive and healthy workplace. <i>(Evaluating, Creating).</i>						
Course Content:							
Unit-1:	Introduction to Modern Office Management 1.1 Definition and Importance of Office Management 1.2 Role and Functions of office staff 1.3 Office Organization and Administration 1.4 Types of Office and Their Characteristics 1.5 Office Ethics and Professionalism	8Hours					
Unit-2:	Communication and Coordination 2.1 Fundamentals of Business Communication 2.2 Types and Channels of Communication 2.3 Barriers to Effective Communication 2.4 Coordination in the Office Environment 2.5 Techniques for Effective Meetings and Presentations 2.6 Report writing: The 5W-H plan for writing 2.7 Steps in writing workplace documents 2.7 Basics of Editing and document layouts	8 Hours					
Unit-3:	Technology and Office Automation	8 Hours					

	3.1 Overview of Office Automation Systems 3.2 Common Office Software (Word Processing, Spreadsheets, Presentation Software) 3.3 Email and Scheduling Tools 3.4 Use of Internet and Intranet in Office Management 3.5 Introduction to Database Management Systems	
Unit-4:	Records Management and Information Retrieval 4.1 Importance of Records Management 4.2 Types of Records and Filing Systems 4.3 Electronic Records Management 4.4 Information Retrieval Systems 4.5 Legal and Ethical Issues in Records Management	8 Hours
Unit-5:	Office Layout and Ergonomics 5.1 Principles of Office Layout 5.2 Space Management and Utilization 5.3 Ergonomics in the Office: Importance and Implementation 5.4 Office Furniture and Equipment 5.5 Health and Safety in the Office staff	8 Hours
Reference Books:	"Office Management: A Productivity and Effectiveness Approach" by David W. Bates. "Modern Office Management" by S.P. Arora. "Principles of Office Management" by R.C. Bhatia. "Business Communication: Building Critical Skills" by Kitty O. Locker and Stephen Kyo Kaczmarek. "Effective Office Management: A Best Practice Guide" by Susan L. Brock. <i>* Latest editions of all the suggested books are recommended.</i>	
Value Added Courses (VAC)		
Course Code TMUVA240101/0 201	Value Added Course -1 BCA Semester -I Indian Knowledge Systems	L-2 T-0 P-0 C-2
Course Outcomes:	On completion of the course, the students will be :	
CO1.	Understanding Indian Civilization, Geography, Natural Resources & Culture.	
CO2.	Understanding Constitution of India	
CO3.	Understanding & applying the contribution of Bharat Varsha in Mathematics & other Sciences to the present age.	
CO4.	Understanding and applying ideals of Indian Philosophy In their life.	
CO5	Understanding origin of Medicine & importance of Environment.	
Course Content:		
Unit-1:	BHARATVARSHA – THE LAND OF CIVILIZATION & CULTURE: The Aryavart, Indus Valley Civilization, Harappa, Mohenjo-Daro, Fine Arts & Dance Forms, Varnashrama System, Education System, Occupations & Rituals. RICH GEOGRAPHY: Mountains, Rivers, Deserts, Coast Line, Soil Types, Natural Resources, Minerals, Seasons & Weathers, Agriculture & Major crops.	6Hours
Unit-2:	BHARATVARSHA – CONSTITUTION OF INDIA: Drafting of	6 Hours

	Constitution, Drafting Committee, Articles & amendments, Fundamental duties, Fundamental Rights, Creation of States & Union Territories, Appointment of The President & Governor, Prime Minister & Chief Minister, Upper House & Lower House in Parliament, Appointment of Supreme Court Judges, High Court Chief Justice & Chief Justice of India.	
Unit-3:	BHARATVARSHA – LAND OF SCIENCE, MATHEMATICS, ASTRONOMY & ASTROLOGY: Vedic Mathematics, Contribution of Ancient Indian Mathematicians, Scientists & thinkers to the modern age of Science, Mysteries of Astronomy and Astrology as per ancient Indian literature	6 Hours
Unit-4:	BHARATVARSHA- VEDAS, UPNISHADS, EPICS – Indian Education System, Languages, Elementary idea of all 4 Vedas & Upanishads & their teachings, Schools of Vedic Philosophy.	6 Hours
Unit-5:	BHARATVARSHA – HEALTH & ENVIRONMENT: Health & Disease, description in Vedas, Ayurveda- Charaka, Surgery- Susruta, Yoga- Patanjali. Importance of Environment & Message To Protect Environment as per Vedas.	6 Hours
Text Book	1. Mahadevan, B., Rajat Bhai, Vinayak & Nagendra Pavana, R.N., "Introduction to Indian Knowledge System". 2. Jha, Amit, "Traditional Knowledge System in India". Pole, Prashant, "Treasure Trove of Indian Knowledge". 3. Chauhan, Bhag Chand, "IKS The Knowledge System of Bharata" (A Textbook Based On NEP 2020 Guidelines). <i>Sen Gupta, Nirmal, "Traditional Knowledge In Modern India"</i>	
Reference Books:	1. Rao, Sheshgiri, K.L., "Encyclopedia of Hinduism" VOL II, VOL III & VOL IV. 2. Kapoor, Kapil, "Encyclopedia of Hinduism". 3. Mishra, Ashok, "Hinduism- Ritual, Reason & Beyond". 4. Vatsyayan, Kapil, "The Cultural Heritage of India"	
Additional Electronic Reference Material	https://en.wikipedia.org/wiki/Geography_of_India https://amritmahotsav.nic.in/blogdetail.htm?69 https://en.wikipedia.org/wiki/Constitution_of_India https://prepp.in/news/e-492-appointment-of-judges-of-high-court-collegium-system-indian-polity-upsc-notes https://timesofindia.indiatimes.com/vedic-perspective-on-environment/articleshow/4613346.cms	
Course Code TMUVA240201	Value Added Course -2 BCA Semester -II Environmental Science & Disaster Management	L-2 T-0 P-0 C-2
Course Outcomes:	On completion of the course, the students will be :	
CO1.	Understanding environmental problems arising due to constructional and Developmental activities.	
CO2.	Understanding the natural resources and suitable methods for conservation of resources for sustainable development.	
CO3.	Understanding the importance of ecosystem and biodiversity and its conservation for maintaining ecological balance.	
CO4.	Understanding the types and adverse effects of various environmental pollutants and their abatement devices.	
CO5	Understanding the general concept in the dimensions of disasters caused by nature beyond the	

	human control as well as the disasters and environmental hazards induced by human activities with emphasis on disaster preparedness, response and recovery, etc.	
	Course Content:	
Unit-1:	Introduction of Environmental Studies: Definition and Scope of environmental studies, multidisciplinary nature of environmental studies, Concept of sustainability & sustainable development. Ecology and Environment: Concept of an Ecosystem-its structure and functions, Energy Flow in an Ecosystem, Food Chain, Food Web, Ecological Pyramid & Ecological succession, Study of following ecosystems: Forest Ecosystem, Grass land Ecosystem & Aquatic Ecosystem & Desert Ecosystem.	6 Hours
Unit-2:	Natural Resources: Renewable & Non-Renewable resources; Land resources and land use change; Land degradation, Soil erosion & desertification. Deforestation: Causes & impacts due to mining, Dam building on forest biodiversity & tribal population. Energy Resources: Renewable & Non-Renewable resources, Energy scenario & use of alternate energy sources, Case studies	6 Hours
Unit-3:	Biodiversity: Hot Spots of Biodiversity in India and World, Conservation, Importance and Factors Responsible for Loss of Biodiversity, Biogeographical Classification of India Environmental Pollutions: Types, Causes, Effects & control; Air, Water, soil & noise pollution, Nuclear hazards & human health risks, Solid waste Management; Control measures of urban & industrial wastes, pollution Climate change & Global Warming (Greenhouse Effect), Ozone Layer -Its Depletion and Control Measures, Photo chemical Smog, Acid Rain	6 Hours
Unit-4:	Environmental Policies & Practices: Environmental laws: Environment protection Act (overview); Air Prevention & Control of Pollution Act (Overview), Water Prevention & Control of Pollution Act (Overview), Wild Life Protection Act (Overview), Forest Conservation Acts (Overview), International Acts; Montreal & Kyoto Protocols & Convention on biological diversity, Nature reserves, tribal population & Rights & human wild life conflicts in Indian context	6 Hours
Unit-5:	Disaster Management: Introduction on Disaster, Definition and types of disaster, Study of Important disasters, Causes, effects and practical examples for all disasters, Risk and Vulnerability Analysis, Disaster Preparedness and Response, Rehabilitation, Reconstruction and Recovery, Mitigation and Management techniques of Disaster, Training, awareness program and project on disaster management.	6 Hours
Text Books:	1. "Environmental Chemistry", De, A. K., New Age Publishers Pvt. Ltd. 2. "Introduction to Environmental Engineering and Science", Masters, G. M., Prentice Hall India Pvt. Ltd. 3. "Fundamentals of Ecology", Odum, E. P., W. B. Saunders Co. 4. Disaster Management Guidelines, GOI-UND Disaster Risk Program (2009-2012) Damon, P. Copola, (2006) Introduction to International Disaster Management, Butterworth Heineman.	
Reference Books:	1. "Biodiversity and Conservation", Bryant, P. J., Hypertext Book 2. Textbook of Environment Studies", Tewari, Khulbe & Tewari I.K. Publication 3. https://www.nationalgeographic.org/encyclopedia/biodiversity *Latest editions of all the suggested books are recommended.	
Course Code	Value Added Course -3	L-2

TMUVA240301	BCA Semester -III Digital Technology in Education	T-0 P-0 C-2
Course Outcomes:	On completion of the course, the students will be :	
CO1.	Understanding the meaning of digital education	
CO2.	Understanding the potential of digital technologies to support learning and teaching	
CO3.	Understanding and identifying different tools of digital education available in India.	
CO4.	Understanding the e-governance Digital India and Initiatives	
CO5	Developing sense of responsibility as citizen in digital growth in India	
Course Content:		
Unit-1:	Introduction to Digital Education Meaning & Evolution of Digital Systems. Role & Significance of Digital Technology, digital education vs traditional education, advantages and disadvantages of digital education.	6 Hours
Unit-2:	Digital Education Tools Information & Communication Technology & Tools Interactive tools- Microsoft Teams, Google Classroom, LinkedIn Creative Tools - Google Slides, Google Spreadsheets, Google form, YouTube)	6 Hours
Unit-3:	Digital Education in India Government initiatives for Digital education in India: SWAYAM, E-Path Shala, National digital library of India (NDL India), Digi- Locker. Advantages & challenges in digital education in India.	6 Hours
Unit-4:	E- Governance Introduction of E-Governance in India, Types of E-Governance- G2C (Government to Citizen), G2E (Government to Employee), G2B (Government to Business), G2G (Government to Government)	6 Hours
Unit-5:	Emerging Technologies in Education - AI and machine learning applications in education. - Block chain technology and its implications for credentialing. - Augmented reality (AR) and mixed reality (MR) in educational contexts.	6 Hours
Text Books:	- TECHNOLOGIES IN EDUCATION By DR G VISVANATHAN DR SK PANNEER SELVAM - Technology in Education (English, Paperback, Maddux Cleborne D) - Essentials of Educational Technology by J C Aggarwal - Digital Education for the 21st Century 1st Edition 2021 Softbound by PAL, Apple - Digital Education In India A Virtual Reality by Vardhan And Peeta Bobby And Dwarapudi Bhanu Priya, Om Publications - Education in the Digital Age:How We Get There By Nadav Zeimer - New Digital Technology in Education: Conceptualizing Professional Learning for Educators by Wan Ng (Author)	
Additional Electronic Reference Material:	www.slideshare.net www.lisportal.com/en/lis-blog	
		L-2

Course Code TMUVA240401	Value Added Course -4 BCA Semester -IV Human Values and Professional Ethics	T-0 P-0 C-2
Course Outcomes:	On completion of the course, the students will be :	
CO1.	Understanding the core values that shape the ethical behavior of a manager in functional areas of an organization.	
CO2.	Understanding the morals & values that ought to guide the management and resolve the moral issues in the profession	
CO3.	Analyzing the role of morals and values in technological development and its challenges.	
CO4.	Developing value based management system and work environment in organization.	
CO5.	Developing framework for management ethics and human Values.	
Course Content:		
Unit-1:	Evolution and introduction of Human Values in Indian context: Values: Introduction, sources of value system, Types of values, Significance of values, Indian value system, values of Indian manager, Teaching from Indian scriptures like Mahabharata, Bhagwad Gita, Ramayana, Quran and its applications in management.	6 Hours
Unit-2:	Business Ethics: Definition of Ethics, nature of ethics, types of ethics, Ethics and morality, Need and significance of business Ethics, code of conduct and Ethics for manager.-	6 Hours
Unit-3:	Ethical Dilemmas: Ethical Dilemmas- sources and their resolutions. Ethical decision making, Work ethics, Ethical and unethical practices in India.	6 Hours
Unit-4:	Impact of Ethics: Ethical issues related with marketing, Human resource management, Finance, Intellectual property rights. Environmental ethics.	6 Hours
Unit-5:	Understanding Harmony in the Family and Society –harmony in Human -Human Relationship and Gender issues: Understanding Harmony in the Family and Society-Harmony in Human-Human Relationship; Understanding harmony in the Family-the basic unit of human interaction; Understanding values in human-human relationship; meaning of Nyaya and program for its fulfillment to ensure Ubhay - tripti; Trust (Vishwas) and Respect (Samman) as the foundational values of relationship. Gender Issues and Biases: Gender Stereotyping and Gender Discrimination, Male Gaze and Objectivity, Remedial measures of gender biasness.	6 Hours
Text Books:	1. Bhatia S.K., <i>Business Ethics and managerial values</i> , New Delhi: <i>Deep and Deep publications Pvt. Ltd.</i>	
Reference Books:	1. Velasquez, <i>Business Ethics – Concepts and Cases</i> , New Delhi: Prentice 2. Mathur U.C., <i>Corporate Governance & Business Ethics</i> , New McMillan. 3. Govindarajan M., <i>professional ethics and Human values</i> , New Delhi learning pvt ltd. 4. Hartman Laura P & Abha Chatterjee, <i>Business Ethics</i> , New Delhi: M Hill. 5. R. K Sharma, puneetagoel, <i>Corporate governance, values and ethic Delhi: kalyani publication</i> <i>* Latest editions of all the suggested books are recommended.</i>	